Canadian Waterways Explorer ??? Content Expansion Spec

Content Sweep and Expansion Specification

Prepared 2026-10-16. This document replaces the earlier specification in full.

How it was produced. Every screen file under `mobile/src/app` and the components they render were read line by line (file:line references below are to those reads). Every backend route the app calls was read in full. Every count and every "filled / not filled" statement was produced by read-only queries against the live Turso database on 2026-10-16 (`sweep-verify.ts`, `sweep-verify2.ts`, kept in the session scratchpad; the row-level output is in `sweep-verify.json`). Nothing in sections 1 to 4 is inferred from `schema.prisma` or from file names. Where something could not be verified in this session it is marked **unverified**. Historical statements in section 5 are domain knowledge offered for the owner's judgement; the two things that section cannot assert without running the tools are marked as such: whether OpenStreetMap holds a usable geometry for a waterway, and whether a verified likeness exists for a person.

Scope rule (owner's instruction). The app is about early North American exploration, trade and cartography as it bears on Canadian history. It is not gated to the present border: New France to the Gulf of Mexico, the Dutch trade on the Hudson, Lake Champlain, Peter Pond, Astor and Astoria, Fort Vancouver, the Spanish Pacific voyages and the Russians all belong. Not everything needs to link to everything; a link is wanted only where the app has a surface that shows it and the connection is historically real.

Vocabulary. "Turso" is the database. `@prisma/client` is the query library the backend uses to reach it; it is not a database. `schema.prisma` is the declaration the client was generated from and is not the source of truth.

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0. Numbers at a glance (live, 2026-10-16)

Table	Rows	Published	Notes
Waterway	70	n/a	River 29, Bay 15, Lake 12, Strait 6, Sound 4, Inlet 3, plus one row typed `lake`; 66 with KML; 69 with image; 59 with Indigenous name
Location	100	n/a	Fort 52, Trading Post 16, Portage 14, Cultural Site 9, Settlement 5, `historic_site` 3, `fort` 1; all 100 have an image
Explorer	92	n/a	70 with portrait; 42 with at least one waterway link; 105 ExplorerWaterway rows, all with year and notes
FurTradeInfo	21	n/a	one per waterway
ArchaeologicalDiscovery	8	n/a	all with coordinates
Cartographer	11	n/a	Bellin ×2, Jolliet ×2; 1 location links to a cartographer
IndigenousNation	14	n/a	7 nations, each stored twice; no screen reads this table
NotableFigure	18	n/a	7 featured; 4 with image

Table	Rows	Published	Notes
LessonPlan	26	26	all have narrative, hero, images, key figures, timeline; `visibility` = private on all 26
VirtualFieldTrip / FieldTripStop	6 / 36	6	every stop has image, fun fact and think question; 0 stops linked to a pin or waterway
PrimarySourceDocument	48	48	34 without grade level; 1 with image; 43 linked to an explorer
Quiz / QuizQuestion	13 / 71	13	6 quizzes exist twice; 16 questions carry `sourceType` = explorer with no `sourceId`
ComparisonTemplate	4	4	
PrintableResource	8	8	all HTML content, all with answer key and teacher notes, none with file or preview image
TimelineEvent	23	n/a	1613 to 1885; 0 images; 4 explorer links, 5 location links, 0 waterway links
HistoricalEvent	48	n/a	1534 to 2016; 31 images; 2 waterway links, 2 location links, 0 explorer links
PronunciationGuide	33	n/a	0 audio; 0 link columns filled
IndigenousWord	298	n/a	10 language labels; 0 linked to a waterway or location; 0 audio
VocabularyQuiz	4	4	
IndigenousStory	3	3	Cree, Ojibwe, Inuit
VoyageurJourney / JourneyNode	2 / 23	2	all nodes have coordinates and a historical fact; 0 cover images
Achievement	28	n/a	12 of them count activities no route ever records (see 2.17)
DailyChallenge	30	n/a	dates 2026-02-11 to 2026-03-12, rotated by day index
WordOfTheDay	4	n/a	created on demand

User and operational tables (contributions 2, teachers 7, classes 2, assignments 5, students 0, journals 0, progress 0, map annotations 0, image health logs 6,150) are not content and are not treated further.

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1. Screen by screen

Each entry gives: the data hook and the route it calls; the fields the screen reads; filters and gating; every cross-record connection and the mechanism that realizes it; and what a record must carry to be complete on that screen. Paths are relative to `mobile/src/` unless they start with `backend/`.

1.1 Student side

Map tab, native: `app/(tabs)/index.tsx`

- **Data.** `useWaterways()` (288) → `GET /api/waterways` (`backend/src/routes/waterways.ts:45-62`: id, name, indigenousName, lat, lng, regionName, type.name). `useLocations()` (290) → `GET /api/locations` (`locations.ts:21-40`: id, name, indigenousName, lat, lng, locationType, yearEstablished, imageUrl, waterway{id,name}). `useExplorers()` (291) → `GET /api/explorers` (`explorers.ts:7-21`).

`useExplorerDetail(selectedExplorerId)` (292) for the explorer filter. `useRiverOverview()` (289) → `GET /api/waterways/overview/rivers`, the committed `generated/river-overview.json` built by `backend/scripts/build-waterway-overview.ts --types=River` (line paths only for line waterways, 172). `useWaterwayGeometry(id, tier, bbox)` (564-565) → `GET /api/waterways/:id/geometry`.

- **Filters.** Legend offers only River, Lake, Bay, Fort, Trading Post, Portage (967-1010); Strait, Sound, Inlet, Settlement, Cultural Site and `historic_site` have no legend entry (471-480, 511-520). Era chips (30-36) filter locations by `yearEstablished` with inclusive bounds (450-455, 537-542) and keep a waterway only if one of its locations falls in the era (447-461, 495-498). Explorer filter keeps the waterways in `explorerDetail.waterways[].waterway.id` and the locations on them (409, 491, 531).
- **Connections.** Explorer → waterway by `ExplorerWaterway` rows (409). Location → waterway by `Location.waterwayId` (458, 524, 531). Overview record → waterway by shared id (427-428). Tap hit-testing on the overview is geometric: bbox prefilter, point-in-ring, 24 px segment distance (202-249).
- **Complete record here.** A waterway needs `latitude`/`longitude`, `type`, and `kmlData` for a trace (overview for rivers, geometry route for all). A location needs coordinates, a `locationType` the legend knows, a `yearEstablished` for the era filter (a null year passes every era, 450), and a `waterwayId`. An explorer needs at least one `ExplorerWaterway` row or the filter returns nothing (490, 530).

Map, web: `app/(tabs)/index.web.tsx`

- Lists waterways and locations (26-27), detail panels via `useWaterwayDetail`/`useLocationDetail` (31-36). Shows explorers (158-165), fur trade (172-194), location's waterway and cartographer (238, 245). No navigation, no filters (166, 170). Reads `regionName` (85), the only screen that does.

Detail sheet: `components/DetailBottomSheet.tsx`

- **Data.** `useWaterwayDetail` (151-153) → `GET /api/waterways/:id` (`waterways.ts:194-264`, includes type, explorers{explorer} by yearExplored, furTradeInfo, locations{cartographer}, discoveries; strips `kmlData`, adds `hasKml`, 253-260). `useLocationDetail` (154-156) → `GET /api/locations/:id` (`locations.ts:72-141`, includes waterway{type, explorers{explorer}, furTradeInfo}, cartographer). Contributions by waterway/location (159-164). `useExplorerDetail(selectedExplorerId)` (169).
- **Fields.** Waterway: type, imageUrl, galleryImages (JSON, parsed 63), videoUrl (JSON array or plain URL, parsed 102-123), description, historicalSignificance, indigenousName/Language, explorers[] with year and notes (564-640), notableFigures[] (653-700), discoveries[] (704-780), furTradeInfo (787-835). Location: locationType, yearEstablished, cartographer.name (946-952), historicalNotes, and the **parent waterway's** explorers, notable figures and fur trade (975-1157), plus a non-tappable "Connected Waterway" card (1166-1179).
- **Connections and mechanisms.**
  - Waterway → explorers: `ExplorerWaterway` rows; tap opens `ExplorerDetailModal` (608, 1375-1378).
  - Waterway → notable figures: **text match**. The server runs `notableFigure.findMany({ where: { associatedLocations: { contains: waterway.name } } })` and filters the parsed array for the exact name (`waterways.ts:217-251`; same on `location.waterway.name` in `locations.ts:96-130`). A location's own name is never matched; a location shows its waterway's figures. Tap → `/notable-figures/:id` (668, 1076).
  - Waterway → discoveries: `ArchaeologicalDiscovery.waterwayId`. The discovery's explorer is `relatedExplorerName`, **text only**, not linked (751-753).
  - Location → cartographer: `cartographerId` (952).

    - Reading guide: **static file** `data/locationReadingGuideData.ts` keyed by record id, used for both waterway ids (841) and location ids (1186).
    - Contribution: `waterwayId`/`locationId` passed to `ContributeModal` (1361-1364).
- **Complete record here.** Waterway: description, historicalSignificance, imageUrl, 3+ gallery entries `{url, caption, credit}`, explorers with `yearExplored` and `expeditionNotes` (year shows "N/A" otherwise, 578-596), fur trade where it applies, and a reading-guide entry. A notable figure appears on a waterway only if `associatedLocations` contains the waterway's exact `name`. Location: yearEstablished, historicalNotes, image and gallery, cartographer where one surveyed it, reading-guide entry.

***Explorer modal:* `components/ExplorerDetailModal.tsx`**

- Props only (18-22). Reads imageUrl, name, nationality, birth/death (shown only when both exist, 46), biography, notableAchievements, waterways[] with type badge, year and notes (72-110). Waterway rows are not tappable (78-81). Reading guide from static `data/readingGuideData.ts` keyed by explorer id (117, 123).

***Explorers tab:* `app/(tabs)/two.tsx`**

- `useExplorers()` (40); sorted by the last whitespace-separated word of `name` (20-28, 44-51), so "Sieur de La Vérendrye" sorts under "Vérendrye" and "Captain George Vancouver" under "Vancouver". Card shows portrait, name, nationality verbatim, years (71-84). Tap → modal (68, 153-156). No filter.

***Quizzes tab:* `app/(tabs)/quizzes.tsx` *and* `app/quiz/[id].tsx`**

- `useQuizzes(filters)` (188-190) → `GET /api/quizzes` (isPublished; `quizzes.ts`). Category chips hard-coded: Explorers, Fur Trade, Indigenous Heritage, Geography, Maritime History (19); grade chips K-3, 4-6, 7-9, 10-12 (23). Filters applied server-side and again client-side: category by substring (200), grade by equality (207). Estimated time = question count (81).
- Quiz screen posts `{userId, answers[{questionId, selectedAnswer}]}` to `POST /api/quizzes/:id/attempt` (336-345; `education-api.ts:1571-1573`); the server grades and awards points via `completeResource` (`quizzes.ts:237-256`). Question `sourceType`/`sourceId` are not read by this screen.
- **Complete record here.** title, description, category from the hard-coded five, gradeLevel from the four bands, difficulty, 5+ questions with options, correctAnswer id, explanation.

***Deep Dives:* `app/lessons/index.tsx`, `app/lessons/[id].tsx`**

- List: `useLessonPlans({gradeLevel, topic})` (41-44) → `GET /api/lesson-plans` (`lesson-plans.ts:14-50`). Topic chips are **hard-coded** (25-34): Fur Trade, Geography, Indigenous Routes, Maritime Discovery, Metis History, Women in History, Cartography, Early Explorers. Live topics match this list exactly.
- Detail: `useLessonPlan` (110) → `GET /api/lesson-plans/:id` (`lesson-plans.ts:53-108`, parses images, keyFigures, timeline, curriculumConnections, objectives, materials, activities, discussionQuestions, teacher* arrays, related*Ids). Shows hero, grade, reading time, title, description, topic, `narrativeContent` (197-200), key figures (207-223), timeline (226-262), gallery (266-287), Mark Complete (290).
- **Connections and mechanisms.**
    - Key figure → explorer: `keyFigures[].explorerId` inside the JSON, read through `as any` (67, 213); the portrait is taken from the explorer record first (`explorerData?.imageUrl ?? figure.imageUrl`, 71); tap opens `ExplorerDetailModal` (218, 296-298). Live: 105 of 105 key figures carry an `explorerId` that resolves.
    - Timeline item → `TimelineEvent`: `timeline[].eventId` inside the JSON (233) → `useTimelineEvent` (115) → `GET /api/timeline/:id`. Items without `eventId` are pressable but open nothing (234-236). Live:

68 of 123 timeline items resolve.

  - `relatedWaterwayIds`, `relatedLocationIds`, `relatedExplorerIds`: parsed by the server, **read by no screen**. Live: empty on all 26.
  - Completion: `POST /api/lesson-plans/:id/complete` with `{userId}` (`MarkCompleteButton.tsx:53-56`, `education-api.ts:643`), 15 points, assignment completion (`lib/progress.ts`).
- **Complete record here.** gradeLevel, topic from the chip list, heroImageUrl, narrativeContent, images[] with caption and credit, keyFigures[] each with `explorerId` (and the person present in `Explorer` with a portrait if a likeness exists), timeline[] each with `eventId` pointing at a real `TimelineEvent`, readingTimeMinutes (live: 5 of 26 set).

Timeline: `app/timeline.tsx`

- `useTimelineEvents({theme, importance})` (75-78) → `GET /api/timeline` (`timeline.ts:7-61`); themes from `GET /api/timeline/themes`. Grouped by century in arrival order, assumes server year order (88-97). `explorerId`/`waterwayId`/`locationId` are shown as the words "Explorer", "Waterway", "Location" (174-191) and in the modal as "Related …" labels (440-457); **nothing is fetched or navigable**. No images are read.
- **Complete record here.** year (+ month/day/endYear/isApproximate), theme from the live set (exploration, fur_trade, indigenous, settlement), importance (major/moderate/minor), description. Link ids only matter if a future change makes them navigable.

Field Trips: `app/field-trips/index.tsx`, `app/field-trips/[id].tsx`

- List: `useFieldTrips({gradeLevel, theme})` (30-33) → `GET /api/field-trips` (`field-trips.ts:14-64`). Grade chips from `GRADE_LEVEL_RANGES`; `selectedTheme` exists but has **no chip UI** (28, 32).
- Detail: `useFieldTrip` (38) → `GET /api/field-trips/:id` (stops ordered, `field-trips.ts:67-87`). The map region is the average of stop coordinates (85-96); stops without coordinates are dropped from the map (82-84). Shows title, image, description, narrative, audio button with no handler (262-267). Introduction on first stop, conclusion and Mark Complete on last (214, 272, 280-281) → `POST /api/field-trips/:id/complete`, 20 points.
- **Connections.** None to other records. `FieldTripStop.locationId`/`waterwayId` exist in the admin schema (`backend/src/routes/admin-field-trips.ts:33-45`) and are 0 of 36 filled; the screen does not read them. Stop titles are prose ("David Thompson's Base Camp"); no stop title equals a location or waterway name, and the trips' subjects (Fort Carlton, Fort William, Prince of Wales Fort, Rocky Mountain House, York Factory) exist as `Location` rows only by **coordinate coincidence**, not by any join.
- **Complete record here.** gradeLevel, theme (live values `fur_trade`, `explorer`), estimatedMinutes, coverImageUrl, 6 stops each with coordinates, image, description, funFact, thinkQuestion. The screen also reads `stop.narrative` (255-258), which is neither in the admin stop schema (`admin-field-trips.ts:33-45`) nor a column in Turso (2026-10-14 introspection), so that section never renders; `audioUrl` is a column and is null on all 36 stops.

Documents: `app/documents/index.tsx`, `app/documents/[id].tsx`

- List: `useDocuments({type})` (38-40) → `GET /api/documents` (isPublished, filters type and gradeLevel, `documents.ts:7-43`). Type chips from `GET /api/documents/types` (150). **There is no grade filter UI**; `gradeBadge` styles are unused (302-311). Badge colour map keys: map, journal, letter, treaty, photograph, artwork, newspaper (24-32); live types are journal, report, letter, inscription, map, so `report` and `inscription` take the grey fallback.
- Detail: `useDocument` (49) → parses annotations, vocabulary, discussionQuestions (`documents.ts:59-84`). Shows image, type, grade, author, date, historicalContext, originalText and transcription, annotations by grade, vocabulary, discussion questions (80-205).

- **Connections.** `explorerId`/`waterwayId`/`locationId` are returned raw and **not read by either screen**. The author is a free-text string; 40 of 48 author strings equal an `Explorer.name` exactly.
- **Complete record here.** documentType, author, originalYear (originalDate optional), transcription (originalText where the source is in English), historicalContext, vocabulary[], discussionQuestions[], annotations[] by grade, **gradeLevel** (34 of 48 missing, so they never answer a grade query), imageUrl of the manuscript or a period image (1 of 48).

What Happened Here: `app/nearby-history.tsx`

- `useNearbyHistoryEventsByCategory(lat, lng, radius, category)` (339-349) → `GET /api/nearby-history/by-category` and `GET /api/nearby-history` (haversine radius, `nearby-history.ts:213-250`). Radius 5/10/25/50/100 km, default 50 (62, 328). Category chips are the hard-coded seven: exploration, fur_trade, indigenous, settlement, battle, treaty, maritime (65-101); `maritime` has 0 events. The categories hook result is ignored (336). Sorted by server-computed distance (407-409). "View on Map" pushes `/?lat&lng&eventId` (403).
- **Connections.** By coordinates only. The detail route joins related explorer/waterway/location (`nearby-history.ts:254-314`) but the screen never calls it.
- **Complete record here.** precise latitude/longitude, locationName, year, category from the seven, significance, description, imageUrl (31 of 48). Events in London and Greenhithe (1821 merger, 1845 departure) can only ever be found by a user standing in England.

Notable Figures: `app/notable-figures/index.tsx`, `app/notable-figures/[id].tsx`

- List: `useNotableFigures({figureType, nation})` (235-238) → `GET /api/notable-figures` (`notable-figures.ts:10-66`); featured via `/featured`; types and nations via `/types`, `/nations` (distinct values). `figureTypeConfig` knows woman, indigenous_leader, guide, interpreter, trader (43-74); live types also include `metis_leader` (3) and `voyageur` (1), which take the grey fallback (76-85).
- Detail: `useNotableFigure` (100) → parses associatedExplorers, achievements, associatedLocations (`notable-figures.ts:148-175`). Associated explorers render as cards with a chevron but are plain `View`s with **no press handler** and no id (255-265). Associated locations are text chips (277-281). Biography and significance render without a null guard (219-230).
- **Connections.** Figure → waterway is the text match in 1.1 Detail sheet (server side); figure → explorer is `{name, relationship}` text; 12 of 23 associated-explorer names equal an `Explorer.name` exactly, 11 do not (John Franklin vs "Sir John Franklin", John Richardson, George Back vs "Sir George Back", each twice; Lord Selkirk, Thomas Scott, Meriwether Lewis, William Clark, Toussaint Charbonneau).
- **Complete record here.** figureType from a canonical set, nation, role, biography, significance, achievements[], associatedExplorers[] with names spelled exactly as in `Explorer`, associatedLocations[] containing the exact `Waterway.name` strings for the waterways the figure should appear on, birth/death or active period, isFeatured, imageUrl only where a likeness exists.

Indigenous Languages: `app/indigenous-languages.tsx`

- `useWordOfTheDay`, `useIndigenousLanguages`, `useIndigenousWords({language})`, `useVocabularyQuizzes`, `useIndigenousStories` (65-71) → `/api/indigenous-language/*` (`indigenous-language.ts`). Language tabs from the live groupBy. Word cards show language, category, word, phonetic, translation, culturalContext (185-214).
- **Quiz cards (226) and story cards (251) are plain `View`s, not pressable.** Detail hooks `useVocabularyQuiz`, `useSubmitVocabularyQuiz`, `useIndigenousStory` exist (`education-api.ts:911-953`) and routes exist, but no screen opens a story or plays a vocabulary quiz. Colour map keys Cree, Ojibwe, Inuktitut, Mi'kmaq, Dene, Multiple

(50-57); Blackfoot, Mohawk, Iroquois fall to the default; story colour uses the story's **nation** against the language map (249).

- **Connections.** None rendered. `relatedWaterwayId` / `relatedLocationId` on words and stories are 0 filled and unread.
- **Complete record here.** Word: word, translation, language, phonetic, culturalContext, category. Story: title, nation, language, content, summary, theme, gradeLevel. Vocabulary quiz: title, language, difficulty, questions[]. A story or quiz is "complete" only once a screen can open it.

***Pronunciations:* `app/pronunciations.tsx`**

- `usePronunciations({termType, language})` (41-44), term types and languages from the server. Shows term, phonetic, meaning, play button only with `audioUrl` (106). Live: 0 of 33 have audio. Colour map keys waterway, location, nation, explorer, general (28-34); live termTypes are indigenous_nation, indigenous_place, french, explorer, so three of four take the fallback. No connections rendered; link columns 0 filled.

***Compare:* `app/comparisons/index.tsx`, `app/comparisons/[id].tsx`**

- `useComparisons({type, gradeLevel})` (37-40), `useComparison` (49) → `ComparisonTemplate` only. Items are free-form JSON; the label key is the first of name, perspective, route, title (21-32); every key/value is rendered (133-158). The generated comparisons (`/api/comparisons/generate/explorers`, `/generate/forts`; `comparisons.ts:84-147, 165-196`) have hooks (`education-api.ts:333, 343`) but **no screen calls them**. Colour keys explorers, forts, waterways, nations, trading_companies (24-30); live types are explorers, perspectives, trade_companies, trade_routes, so three of four take the fallback.

***Printables:* `app/printables/index.tsx`, `app/printables/[id].tsx`**

- `usePrintables({type, topic, gradeLevel})` (61-65); topics are fetched but there is no topic UI (58, 67). Detail renders `content` HTML in a WebView (50, 121); Download only with `fileUrl` (143), which no row has. Public route strips answerKey and teacherNotes (`printables.ts:91-94`). Colour keys worksheet, map, timeline, activity_sheet, quiz_printable match live.

***Voyageur Journeys:* `app/voyageur-journey/index.tsx`, `app/voyageur-journey/[id].tsx`**

- `useVoyageurJourneys` (167), `useVoyageurJourney`, `useStartJourney`, `useMakeJourneyChoice`, `useJourneyProgress` (190-193). No client filter. Choices are posted to the server, which returns the next node (`voyageur-journey.ts`; screen 229-241). Map renders when the node has coordinates (346). `nodeType`, `challengeQuestion`, `challengeAnswer` are not read by the screen. Journey completion awards no `UserProgress` points (route adds `pointsChange` to journey progress only).
- **Connections.** `locationName` is text; in "The Grand Portage Route" two node names equal `Location.name` (Grand Portage, Fort William) by coincidence; nothing joins.

***Achievements:* `app/achievements.tsx`**

- `useUserProgress`, `useAchievements`, `useUserAchievements`, `useDailyChallenge`, `useSubmitDailyChallenge` (413-419). Earned status matches `userAchievement.id` to `achievement.id`; the server returns the achievement's id in `id` (`gamification.ts:181-182`), so the match is correct. Daily challenge: `GET /api/gamification/daily-challenge` rotates the 30-row pool by day index (`gamification.ts:214-283`); submit awards `pointsReward` directly (292-407).
- **Criteria that can never be met.** `recordActivity` is called only for journal entries (`journals.ts:186`) and by `completeResource` for quizzes, lessons and field trips (`lib/progress.ts:267`). The only route that records

`waterway_explored`, `location_visited`, `pronunciation_learned` or `document_read` is `POST /api/gamification/progress/add-points` (`gamification.ts:111-130`), and its client hook (`education-api.ts:993`) is called by **no screen**. So the 12 achievements with those criteria (First Steps, Waterway Wanderer, River Runner, Master Cartographer, Location Scout, Fort Finder, First Discovery, Document Detective, Archive Master, First Words, Language Learner, Polyglot) cannot be earned today.

My Class, Join Class, Journal, My Maps

- `app/my-class.tsx`: `GET /api/classes/mine?userId` (`classes.ts:152-213`); assignment rows route to `/quiz/:id`, `/lessons/:id`, `/field-trips/:id` by `quizId`/`lessonPlanId`/`fieldTripId` (42-47). These are the only content types a teacher can assign (`classes.ts:49-57`).
- Journal: entries may carry `waterwayId`, `locationId`, `explorerId`, `quizId`, `lessonPlanId`, `fieldTripId` (`journals.ts:17-27`) but the entry screen posts only title, content, tags (`journal/entry/[id].tsx:72-90`). Saved locations exist server-side (`journals.ts:264-326`) and no screen calls them.
- My Maps: pins may carry `linkedWaterwayId`/`linkedLocationId`/`linkedExplorerId` and routes `compareToExplorerId` (`map-annotations.ts:31-51`); the screen posts coordinates, title, description, colour only (`my-maps/[id].tsx:236-242`).

Learn hub, settings, onboarding

- `app/(tabs)/learn.tsx` is static navigation (130-323). `settings/*` and `onboarding.tsx` carry no records; onboarding hard-codes "Learn 298 words from 8 Indigenous languages" (71), which is the live word count today and will drift.

1.2 Teacher side

Lesson Plans: `app/teacher/lesson-plans.tsx`, `app/teacher/lesson-plan/[id].tsx`

- Same list endpoint as Deep Dives; same hard-coded topic list (24-34). Detail reads the **teacher face**: objectives, materials, activities `{title, duration, description, materials[]}`, discussionQuestions, `assessmentSuggestions` (192-198), curriculumConnections `{subject, strand, expectation}`, relatedQuizIds as a count (227-236) whose button pushes the generic `/(tabs)/quizzes` with no filter (232).
- **Bug.** The screen reads `lesson.assessmentSuggestions`; the server spreads the row and the column is `assessment` (`lesson-plans.ts:53-108`; schema line 277). The type file declares `assessmentSuggestions` (`lib/types/education.ts:63`). The Assessment section never renders for the 25 plans that have one.

Deep Dives (teacher view): `app/teacher/deep-dives.tsx`, `app/teacher/deep-dive/[id].tsx`

- All plans, no filter (26, 81). Detail merges base and teacher arrays without distinction (91-106); "teacher only" sections are styled, not gated, and the screen has no session check (205, 219). Renders narrativeContent, mainContent, teacherNotes, assessmentSuggestions as Markdown via `components/MarkdownContent.tsx` (H2/H3/H4, bullets, numbered as bullets, bold, italic; no links, images or tables, 51-160). "Assign" pushes `/teacher/assignment/new?lessonId=` (340).

Classes and assignments: `app/teacher/dashboard.tsx`, `class/new.tsx`, `class/[id].tsx`, `assignment/new.tsx`

- Dashboard requires a teacher session (56-59). Class detail shows students and assignments with "x of N students completed" against the current roster (231-232); assignment rows are not pressable and never read the resource ids (205). New assignment picks a quiz, lesson plan or field trip by title substring (144-148), **not filtered by the class's**

grade, and posts `{title, assignmentType, instructions, dueDate, quizId|lessonPlanId|fieldTripId}` (189-200; `classes.ts:603-664`).

1.3 Admin side

Screen	What it can do	Evidence
`(tabs)/admin.tsx`	login, counts, links; admin tab is always declared in the tab bar (`(tabs)/_layout.tsx:105-118`)	101-111, 279-366
`admin/quizzes.tsx`, `quiz-editor/new.tsx`, `quiz-editor/[id].tsx`, `components/QuestionEditorModal.tsx`	full quiz and question editing; question may link `sourceType` explorer/waterway/location + `sourceId` from live lists (QuestionEditorModal 145-147, 319-330); server validates correctAnswer ∈ options (`admin-quizzes.ts:348-360`)	
`admin/lesson-plans.tsx`	list and delete only; Create and Edit show "Coming Soon" (152-160) although `POST/PATCH /api/admin/lesson-plans` exist	
`admin/documents.tsx`	list and delete; Create/Edit "Coming Soon" (167-175); API exists with annotations/vocabulary schemas (`admin-documents.ts:13-60`)	
`admin/field-trips.tsx`	list and delete; Create/Edit "Coming Soon" (170-178); API exists incl. stops with `locationId`/`waterwayId` (`admin-field-trips.ts:33-59, 207-309`)	
`admin/printables.tsx`	list and delete; Create/Edit "Coming Soon" (166-174)	
`admin/contributions.tsx`	approve/reject pending contributions (156-187)	
`admin/my-content.tsx`	placeholder, all counts hard-coded 0 (66-68, 138-170)	
`admin/users.tsx`, `approvals.tsx`, `teacher-approvals.tsx`	account management	
Notable figures	`POST/PATCH/DELETE /api/notable-figures` behind `requireAdmin` (`notable-figures.ts:178-297`); **no screen**	
KML	`POST/DELETE /api/waterways/:id/kml` super admin; `backend/src/admin-panel.html` served at `/admin/kml` (`index.ts:80-83`)	
Image health	`scheduleImageHealthCheck()` daily 02:00 (`jobs/image-health-check.ts:414-431`), covers imageUrl/gallery/hero/cover fields on 14 models (50-171), auto-repairs broken Wikimedia URLs on non-UGC rows (225-264); `admin-image-health.ts` dashboard route	

No editor of any kind (screen or admin route) exists for: Waterway rows (other than KML), Location, Explorer, ExplorerWaterway, FurTradeInfo, Cartographer, IndigenousNation, TimelineEvent, HistoricalEvent, IndigenousWord, PronunciationGuide, IndigenousStory, VocabularyQuiz, ComparisonTemplate, VoyageurJourney/JourneyNode, DailyChallenge, Achievement. These are populated only by scripts against Turso.

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2. Type by type: what a complete record is, and how it connects

For each type: the surfaces (from section 1), the connection mechanisms with the live fill, the minimum data for a new record to be useful, and the live defects.

2.1 Waterway

- **Surfaces.** Map pin and trace, preview card, detail sheet, web list, explorer modal (name and type), lesson key-figure modal, fur trade, discoveries.
- **Connections.**

To	Mechanism	Live fill
WaterwayType	`typeId` column	70/70; one row typed `lake` beside 12 `Lake`
Explorer	`ExplorerWaterway` rows (join)	52 waterways have ≥1; 18 have none
Location	`Location.waterwayId` (column on the location)	33 waterways have ≥1 location
FurTradeInfo	1:1 `waterwayId`	21
ArchaeologicalDiscovery	`waterwayId`	6 waterways
NotableFigure	**text**: `associatedLocations` contains exact `name`	5 waterways reached (Coppermine 3, Columbia 2, Great Slave Lake 2, Great Bear Lake 1, Mackenzie River (Ontario) 1; the other 65 show no figure)
TimelineEvent / HistoricalEvent	`waterwayId`	0 / 2
LessonPlan	`relatedWaterwayIds` JSON	0
FieldTripStop	`waterwayId`	0
QuizQuestion	`sourceType`=waterway + `sourceId`	0
IndigenousWord / IndigenousStory	`relatedWaterwayId`	0 / 0
PronunciationGuide	`waterwayId`; or term equals name	0; 1 term equals an Indigenous name (Kisiskatchewan)
Reading guide	static key = id	68 of 70 (Hayes River and Kaministiquia River lack one)
Geometry	`kmlData` → overview JSON (rivers) and geometry route	66 of 70; missing: Davis Strait, Georgia Strait, Parry Channel, Mackenzie River (Ontario)
Imagery	`imageUrl` + `galleryImages` JSON + `videoUrl` JSON, all on R2	69 (the missing one is Mackenzie River (Ontario)); 279 gallery entries

- **Minimum useful record.** name (with a disambiguating suffix for namesakes, e.g. "Red Deer River (Alberta)"), type from the six live types, latitude/longitude anchor (the KML cleaning guard measures from it), regionName, description, historicalSignificance, indigenousName + indigenousLanguage where attested, kmlData exported with `backend/scripts/export-waterway-kml.ts` (continuity guard, `--with-surface` for wide rivers) then `build-waterway-overview.ts` if a River, imageUrl and 3-5 gallery entries with caption and credit on R2, at least one

ExplorerWaterway with year and notes, FurTradeInfo where trade ran on it, its forts/posts/portages as Locations (which is what makes the era filter find it), and a reading-guide entry.

- **Defects.** Type `lake`; "Mackenzie River (Ontario)" has no KML, no image, no links and is reached only by the Roderick McKenzie text match; 18 waterways without an explorer (Abitibi, Albany, Assiniboine, Atikokan, Bellot Strait, Chesterfield Inlet, Friendly Cove, Hayes, Lake Erie, Lake Nipigon, Lesser Slave Lake, Moose, Placentia Bay, Reindeer Lake, Saguenay, Saint John, Yukon, plus the Ontario row).

2.2 Location

- **Surfaces.** Map pin (legend type, era), preview, detail sheet (its own fields plus the parent waterway's people), web list, generated fort comparison (unused by any screen).
- **Connections.** waterwayId 100/100; cartographerId 1/100 (Fort Témiscamingue → Louis Jolliet); reached from field trip stops 0/36, timeline 5/23, historical events 2/48, documents 1/48, lesson relatedLocationIds 0, quiz source 0, words 0, pronunciation 0; reading guide 83/100 (17 forts lack one, listed in `sweep-verify.json`); image 100/100, gallery 99/100 (350 entries, 1 on commons.wikimedia.org and broken per the last health run), video 100/100.
- **Minimum useful record.** name, locationType from the canonical set (Fort, Trading Post, Portage, Settlement, Cultural Site), latitude/longitude, yearEstablished (null passes every era filter), description, historicalNotes, waterwayId, imageUrl + gallery, indigenousName where attested, cartographerId where a mapmaker surveyed it, reading-guide entry. The locationType must be one the legend knows if it should be filterable (Settlement and Cultural Site are not).
- **Defects.** `fort` ×1 and `historic_site` ×3 beside the canonical labels; Fort William / Fort William (NWC), Fort Edmonton / Fort Edmonton (HBC), Fort Churchill / Fort Churchill (HBC), Fort Garry / Fort Garry (Upper) / Upper Fort Garry, Rocky Mountain House / Rocky Mountain House (NWC) are pairs that need a keep/merge decision; Fort Churchill (HBC, 1783) sits on "Churchill River (Saskatchewan)" and Prince of Wales Fort (1717) likewise, though the Churchill mouth is the Manitoba Churchill (data correctness to confirm, unverified against sources).

2.3 Explorer

- **Surfaces.** Explorers tab, explorer modal, map explorer filter, detail-sheet explorer rows, lesson key-figure cards and modal, generated explorer comparison (unused).
- **Connections.** ExplorerWaterway: 42 of 92 have ≥1; 50 have none and are therefore invisible on the map and show an empty "Waterways Explored". Documents by `explorerId`: 43 docs over 17 explorers. Timeline `explorerId` 4; historical events 0. Lesson key figures by `explorerId` 105 entries pointing at 39 distinct explorers; 53 explorers appear in no lesson. Notable figures naming the explorer by text: 10. Discoveries by `relatedExplorerName`: 6. Pronunciation: 3 terms typed explorer (Des Groseilliers, La Verendrye, Radisson) but the term is not the full name so no screen can match it. Reading guide 92/92. Portrait 70/92; all on R2.
- **Minimum useful record.** name (the surname-sort rule means titles like "Captain", "Sir", "Dr." are harmless but "de La Vérendrye" sorts under V), birthYear/deathYear (both or the lifespan is hidden), nationality from a controlled list, biography, notableAchievements, imageUrl only with a verified likeness, ≥1 ExplorerWaterway with yearExplored and expeditionNotes, reading-guide entry, and where the person wrote: documents with `explorerId`.
- **Defects.** 37 nationality variants printed verbatim ("American (Vermont), Loyalist", "Danish (sailing for Russia)"). 22 without portrait: Alexander Henry the Younger, Charlotte Small, Duncan McGillivray, François de La Vérendrye, Isabel Gunn, Jacques de Noyon, James Knight, Jean-Baptiste Gaultier de La Vérendrye, John McDonald of Garth, Joseph Colen, Louis-Joseph Gaultier de La Vérendrye, Matthew Cocking, Moses Norton, Philip Turnor, Pierre Gaultier de La Vérendrye fils, Richard Norton, Richard Stuart, Robert Bylot, Roderick Mackenzie, Thomas James, William Stuart, William Tomison. Richard Stuart and Robert Bylot have no years. **Fourteen rows are not explorers**

and also exist as NotableFigure: Akaitcho, Big Bear, Charlotte Small, Chief Peguis (figure absent), Crowfoot, Cuthbert Grant, Gabriel Dumont, Isabel Gunn, Jean-Baptiste Lagimodiere, Louis Riel, Marie-Anne Gaboury, Matonabbee, Poundmaker, Thanadelthur (Shawnadithit is in Explorer only). They are there because lesson key figures can only point at `Explorer` (2.9). A rule is needed; see section 6.

2.4 Notable figure

- **Surfaces.** Notable Figures list and detail; detail sheet of the waterways named in `associatedLocations`.
- **Connections.** Waterway by text (6 of 18 figures reach any waterway: Akaitcho, Charlotte Small, Matonabbee, Roderick McKenzie, Sacagawea, Tattannoeuck); explorer by `{name, relationship}` text, no id; lesson key figures reach these people only through their duplicate `Explorer` row. Image 4/18 (3 on R2, Crowfoot on Wikimedia and broken in the last health run).
- **Minimum useful record.** As 1.1 Notable Figures. For the figure to appear on a waterway, `associatedLocations` must hold that waterway's exact `name`; for a location, the location's waterway name (a location name alone does nothing).
- **Defects.** `nation` is free text with 11 variants and does not reference `IndigenousNation`; 9 of the 11 are absent from that table. figureTypes `metis_leader` and `voyageur` have no colour/label config. Associated-explorer names mismatch `Explorer.name` in 11 of 23 entries. Sacagawea has no link into the app other than "Columbia River" (owner's earlier call to keep).

2.5 Indigenous nation

- **Surfaces.** None. `useIndigenousNations()` exists (`lib/api/waterways-api.ts:152-155`) and `/api/indigenous/nations` is served, but no file under `app/` calls the hook. The figure filter "nation" and story `nation` are free-text distinct values, not this table.
- **Connections.** None by column anywhere. Pronunciation terms equal 3 nation names (Cree, Dene, Mi'kmaq).
- **Minimum useful record.** name, alternateNames, languageFamily, traditionalTerritory, culturalNotes. Useful only once a surface exists or `NotableFigure.nation` / `IndigenousStory.nation` / `IndigenousWord.language` are tied to it.
- **Defects.** All 7 nations duplicated (14 rows).

2.6 Cartographer

- **Surfaces.** `locationDetail.cartographer.name` only (DetailBottomSheet 946-952; web 245). No list, no detail.
- **Connections.** `Location.cartographerId`: 1 of 100. Five of the eleven (Thompson, Jolliet, Fidler, Turnor, Champlain) duplicate `Explorer` rows by name.
- **Defects.** Bellin ×2, Jolliet ×2 (one of the Jolliet rows holds the single location link).

2.7 Fur trade info

- 1:1 with waterway; shown in detail sheet and web detail and inherited by locations. 21 rows. `tradingCompany` is free text in 11 spellings of the same four companies plus "Pacific Fur Company", "XY Company", "British maritime traders".

2.8 Archaeological discovery

- 8 rows, shown only in the waterway detail sheet. Explorer by `relatedExplorerName` text (6 of 8 equal an `Explorer.name`; Red Bay has none). Route `/api/discoveries` exists but the app reads discoveries only through

the waterway detail include.

2.9 Lesson plan / Deep Dive (one row, two faces)

- **Surfaces.** Deep Dives (student), teacher Lesson Plans, teacher Deep Dives, assignable.
- **Connections.** keyFigures[].explorerId → `Explorer` 105/105 resolve (JSON id, not a column); timeline[].eventId → `TimelineEvent` 68/123; related*Ids 0/26 and unread; relatedQuizIds 0 and read only as a count. Hero and images 26/26 on R2. Completion → points and assignments.
- **Minimum useful record.** Student face: title, description, gradeLevel (one of four bands), topic (from the hard-coded chip list), heroImageUrl, narrativeContent (markdown; live range roughly 2,500-4,000 characters), images[] `{url, caption, credit}`, keyFigures[] `{name, role, years, description, explorerId}`, timeline[] `{year, title, description, eventId}`, readingTimeMinutes. Teacher face: curriculumConnections[] `{subject, strand, expectation}`, objectives[], estimatedMinutes, materials[], mainContent, activities[] `{title, duration, description, materials[]}`, discussionQuestions[], assessment, teacherObjectives[], teacherActivities[], teacherQuestions[], teacherNotes. Flags isPublished (the public route filters on isPublished only; `visibility` and `isCore` are consulted only by the admin edit-permission helper in `backend/src/lib/admin-middleware.ts:161-190`).
- **Consequence for people.** Because the key-figure link is `explorerId`, every person a Deep Dive wants to make tappable must be an `Explorer` row. That is why Thanadelthur, Riel and the chiefs are in that table. Either the key-figure link gains a `notableFigureId` alternative (code change) or the rule stays "anyone who is a key figure is an Explorer" and the Explorers tab accepts that.

2.10 Virtual field trip and stops

- Covered in 1.1. 6 trips, 36 stops, all stops imaged; 0 stop links; themes `fur_trade`/`explorer`; grades 4-6 ×4, 7-9 ×2; none for K-3 or 10-12. Subjects: Fort Carlton, Fort William, L'Anse aux Meadows, Prince of Wales Fort, Rocky Mountain House, York Factory. Four of the six subjects exist as `Location` rows at the same coordinates with no join; L'Anse aux Meadows exists only as a discovery row.

2.11 Quiz and questions

- 13 quizzes, 6 duplicated by title and content (Canada's Waterways Geography, Explorers of Canada's Coasts, Famous Canadian Explorers, Indigenous Peoples & Canadian Waterways, Lost & Found, The Fur Trade Era); "Explorers of Canada" has 1 question. 71 questions, all multiple choice with explanation. `sourceType`=explorer on 16 with `sourceId` null; the editor can set both (QuestionEditorModal 456-472) and the server stores them, but the editor's reset effect wipes a loaded `sourceId` when `sourceType` changes (235-237). Categories and grade bands are hard-coded in the tab.
- **Minimum useful record.** 5-8 questions; category from the five; gradeLevel from the four; difficulty; sourceType/sourceId where the question is about one record (only useful once a screen reads it).

2.12 Primary source document

- 48 rows: journal 25, report 12, letter 9, inscription 1, map 1. Authors: 17 explorers (Mackenzie 3, Back 3, Cartier 3, Champlain 3, Fidler 3, La Pérouse 4, La Salle 3, Norton 3, R. Stuart 3, W. Stuart 3, Turnor 3, Tyrrell 3, Hearne 3, Thompson 2, Rae 1, Fraser 1, Pond 1) plus Joutel, Knight, Matonabbee, Marten, Saukamappee-via-Fidler. 43 have `explorerId`; the 5 without: Rae's report, Knight on Thanadelthur, Pond's map, Fidler recording Saukamappee, York Factory Trade Record (the last is the only one with waterwayId and locationId). 34 have no gradeLevel; 14 have annotations; 1 has an image.

- **Minimum useful record.** As 1.1 Documents, with gradeLevel always set and explorerId set when the author or subject is an explorer.

2.13 Timeline event

- 23 rows, 1613-1885, all without image; 4 explorer and 5 location links, 0 waterway links. Reached from lessons by eventId: 68 of 123 lesson timeline items carry one, pointing at 18 distinct events; 5 events are referenced by no lesson. Link ids are displayed as labels only.

2.14 Historical event (What Happened Here)

- 48 rows, 1534-2016, GPS-driven. 31 images (29 R2, 2 Wikimedia; the Crowfoot one broken). 2 waterway links, 2 location links, 0 explorer links. Two events are placed in England.

2.15 Indigenous language content

- Words 298 in 10 labels (Cree 83, Ojibwe 58, Inuktitut 45, Dene 34, Blackfoot 30, Mohawk 22, Mi'kmaq 19, Iroquois 5, Seneca/Iroquois 1, Mohawk/Iroquois 1), 9 categories, all with phonetic and cultural context, 0 audio, 0 links. Stories 3, vocabulary quizzes 4, neither openable from any screen. Pronunciations 33, 0 audio, 0 links.

2.16 Comparison, printable, voyageur journey, daily challenge

- Comparisons 4 (explorers, perspectives, trade_companies, trade_routes), items with parallel keys; grades 7-9 ×3, 10-12 ×1. Printables 8, all HTML, grades 4-6 ×4, 7-9 ×3, 10-12 ×1, none K-3. Journeys 2 (Race to the Pacific 1793, 7-9; The Grand Portage Route 1790, 4-6), 23 nodes all with coordinates and facts, 0 cover images, 0 challenges. Daily challenges 30 (explorer 8, indigenous 8, fur_trade 6, waterway 4, voyageur 4).

2.17 Achievements and points

- 28 achievements in 6 categories. Reachable today: total_points (3), daily_streak (3), quizzes_completed (4), field_trips_completed (3), journal_entries (3). Unreachable (12): waterways_explored (4), locations_visited (2), documents_read (3), pronunciations_learned (3), because no screen records those activities (1.1 Achievements). This is a code gap, not a content gap, and it bears on what "complete" means for waterways, documents and pronunciations.

2.18 Static data files (shipped in the app binary)

- `data/locationReadingGuideData.ts`: 155 keys; 68 resolve to waterways, 83 to locations, 4 resolve to nothing (`cmlhbjrrri0017m2mbfts7d8is`, `cmlhcpxjz0007m2a3yptljq2u`, `cmlhcpxk7000dm2a3aywic3h2`, `cmlhcpxka000dm2a3aywic3h2`; two look like typos of neighbouring ids). Entries are `{authorSource, title, publisher?, year?, url?, description?}`; some `authorSource` strings are corrupted by escaping ("Hudson\\", "Lady Franklin\\").
- `data/readingGuideData.ts`: 92 keys, all 92 explorers covered.
- Any new waterway, location or explorer needs an entry here and therefore **an app build**, unlike every other content type, which is server-side.

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3. Connection mechanisms, consolidated

#	Mechanism	Where	Fill today	Note
A	Foreign-key column, joined server-side	Location.waterwayId; Location.cartographerId; FurTradeInfo.waterwayId; ArchaeologicalDiscovery.waterwayId; FieldTripStop.tripId; QuizQuestion.quizId; JourneyNode.journeyId; ClassAssignment.quizId/lessonPlanId/fieldTripId	high where it is the parent key; 1/100 cartographer; 0/36 stop links	the only mechanism the app can navigate reliably
B	Join table	ExplorerWaterway	105 rows, 42/92 explorers	drives map filter and both detail views
C	Id column returned raw, not joined, not read by the screen	TimelineEvent.*Id, HistoricalEvent.*Id (list), PrimarySourceDocument.*Id, IndigenousWord.related*, IndigenousStory.relatedWaterwayId, PronunciationGuide.*Id, LessonPlan.related*Ids, QuizQuestion.sourceId	mostly 0	filling these changes nothing on screen until code reads them
D	Id embedded in a JSON column, read through `as any`	LessonPlan.keyFigures[].explorerId (105/105), LessonPlan.timeline[].eventId (68/123)		the two links that make Deep Dives interactive
E	Text match on a name	NotableFigure.associatedLocations ∋ Waterway.name (server); NotableFigure.associatedExplorers[].name; ArchaeologicalDiscovery.relatedExplorerName; Document.author; JourneyNode.locationName; HistoricalEvent.locationName	partial	breaks silently on any rename or spelling variant
F	Coordinates	FieldTripStop lat/lng; HistoricalEvent radius; JourneyNode lat/lng; map hit-testing	full	coincidence with pins is not a join
G	Static file keyed by id, shipped in the binary	reading guides	155 + 92 keys	needs a build to change
H	Hard-coded taxonomy in the client	quiz categories and grades; lesson topics; nearby-history categories; legend types; colour maps for document type, figure type, pronunciation termType, comparison type, journey difficulty		new values render in grey or are unfilterable until the client changes

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4. Teaching resources by grade band (live)

Band	Deep Dives / Lesson Plans	Quizzes	Field trips	Documents	Printables	Journeys	Comparisons	Stories
K-3	6	2 (1 distinct)	0	0	0	0	0	0

Band	Deep Dives / Lesson Plans	Quizzes	Field trips	Documents	Printables	Journeys	Comparisons	Stories
4-6	7	4 (2 distinct)	4	5	4	1	0	2
7-9	7	5 (3 distinct)	2	9	3	1	3	1
10-12	6	2 (1 distinct)	0	0	1	0	1	0
no band	0	0	0	34	0	0	0	0

Topic coverage of the 26 plans: Fur Trade 12, Indigenous Routes 4, Metis History 3, Cartography 2, Maritime Discovery 2, Early Explorers 1, Geography 1, Women in History 1. Quiz categories: Explorers 5 (3 distinct), Geography 2 (1), Maritime History 2 (1), Indigenous Heritage 2 (1), Fur Trade 2 (1). Nothing on New France, the Pacific voyages, the Arctic searches, cartography, Métis history or women exists as a quiz; no field trip exists outside the fur trade and L'Anse aux Meadows; no document is set for K-3 or 10-12.

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5. Gap analysis in the North American context

The test for inclusion is the owner's: relevant to early North American exploration, trade and cartography as it bears on Canadian history, and useful on at least one surface the app has. For each candidate the surfaces it would serve and the records it would link to are stated. **Geometry availability** means the OpenStreetMap exporter has not been run for it in this session and is unverified until it is. **Likeness** means whether a portrait with a defensible provenance exists; this is unverified for every person named below and must be checked one by one at authoring time (the rule stands: no portrait where none is verified, as with David Thompson).

5.1 Waterways

Present and reasonably connected: St. Lawrence, Ottawa, Saskatchewan, Churchill (Sask.), Columbia, Fraser, Mackenzie, Red, Peace, Athabasca, Lake Athabasca, Lake Winnipeg, Lake Superior, Hudson Bay, James Bay, Nootka Sound, Great Slave Lake, Coppermine, Kaministiquia.

Present but isolated (no explorer, no location, or both): Abitibi, Albany, Assiniboine (3 locations, no explorer), Atikokan, Bellot Strait, Chesterfield Inlet, Friendly Cove, Hayes (the HBC's main road inland from York Factory and the one Thompson, Fidler and every Orkney servant travelled; it has neither explorer nor location nor reading guide), Lake Erie, Lake Nipigon, Lesser Slave Lake, Moose, Placentia Bay, Reindeer Lake, Saguenay (Tadoussac is on the St. Lawrence row), Saint John, Yukon.

Absent, grouped by why they matter to the app's story:

- **Interior of the fur trade (Canada).** North Saskatchewan and South Saskatchewan as named rows (the forts along them sit on the single "Saskatchewan River"); Winnipeg River, Rainy River and Rainy Lake, Pigeon River (the Grand Portage route); Clearwater River with Methye Portage (the portage is a location on the Churchill row); Sturgeon-Weir; Slave River (the link between Athabasca and Great Slave Lake that Mackenzie, Pond and every northern brigade used); Liard; Qu'Appelle; Bow and Oldman; Battle River; Beaver River (Alberta); Lake Manitoba, Lake Winnipegosis, Cedar Lake; Lake Nipissing and the Mattawa (the Champlain/voyageur route to the French River); Lake Timiskaming; Rupert, Eastmain, Severn, Attawapiskat, Missinaibi (the James Bay posts); Churchill

River (Manitoba), which is where Prince of Wales Fort actually stands; Dubawnt, Kazan and the Thelon's neighbours (Tyrrell, Hearne); Hay River; Thompson, Skeena, Stikine, Nass, Kootenay and the Okanagan (New Caledonia and the Columbia Department); Richelieu and Lake Champlain (Champlain 1609, the Iroquois war road, the fur route to Albany); Niagara; Detroit River and Lake St. Clair (Cadillac).

- **New France to the Gulf.** Mississippi (Jolliet and Marquette 1673, La Salle 1682, the Louisiana documents already in the app have nowhere to point); Illinois River and the Chicago portage; Wisconsin and Fox rivers (Marquette's route from Green Bay); Ohio (La Salle, the 1750s contest); Missouri (La Vérendrye's sons 1738-43, Thompson at the Mandan villages 1797, Lewis and Clark); Arkansas and Red River of the South only if the Louisiana story is told.
- **The Dutch and the English colonies.** Hudson River (Hudson 1609, Fort Orange/Albany, the Mohawk trade that pulled furs away from Montreal); Mohawk River; Lake George; Connecticut River is marginal.
- **Pacific and Russian.** Strait of Georgia and Juan de Fuca are present; absent are the Snake River (Ogden, Astorians), Willamette, Puget Sound (Vancouver 1792), Prince William Sound and Cook Inlet (Cook 1778, Russians), Sitka Sound (Baranov, the Russian-American Company), Kodiak, the Alaska coast Bering and Chirikov touched (Lituya Bay is present).
- **Arctic and the search for the Passage.** Davis Strait and Parry Channel exist without KML; absent are Smith Sound, Jones Sound, Wellington Channel (Franklin's last route), Peel Sound, Victoria Strait and Simpson Strait (Rae, Dease and Simpson), Coronation Gulf, Bathurst Inlet, Repulse Bay (Rae's winter base), Wager Bay (Middleton 1742), Roes Welcome Sound (Munk, Knight), Great Fish/Back is present, Hood River, Hudson Strait and Ungava Bay (the whole HBC supply road), Fury and Hecla Strait, Lyon Inlet, Melville Sound.
- **Atlantic approaches.** Cabot Strait, Gulf of St. Lawrence as its own row, Bay of Fundy (Port-Royal is currently pinned to the St. Lawrence row, which is geographically wrong), Annapolis River, Miramichi, Restigouche, Labrador coast (Cartwright, the Moravians), Hamilton Inlet and Churchill River (Labrador), Trinity and Conception bays, Grand Banks only as a bay type if fisheries are told.

Each absent waterway is worth adding only with at least one explorer row, its posts as locations, and a reading-guide entry; a bare trace adds nothing to any screen but the map.

5.2 Locations

Absent forts, posts and places that the existing people and documents already name: Île-à-la-Crosse (Thompson, Charlotte Small, Fidler); Fort Fork (Mackenzie 1793, a document exists); Fort Reliance is present but Fort Enterprise and Fort Franklin are not; Fort Wedderburn and Fort Chipewyan (present) on Lake Athabasca; Pembina (Lagimodière, Gaboury, Gunn); Red River Settlement/Fort Douglas and Seven Oaks (Grant, Riel, a battle event exists); Fort Rouge (an event exists); Fort St. Pierre, Fort St. Charles (present), Fort Maurepas, Fort La Reine, Fort Dauphin, Fort Bourbon, Fort Paskoya (the La Vérendrye chain); Grand Portage is present; Fort Charlotte at its inland end; Rainy Lake House (present as Fort Lac La Pluie); Michilimackinac (an event exists), Sault Ste. Marie (present), Green Bay/La Baye, Fort Frontenac (event exists), Fort Niagara, Fort Detroit (event exists), Fort Pontchartrain; Stadacona and Hochelaga as Indigenous settlements rather than only as Quebec and Montreal; Tadoussac present; Trois-Rivières; Lachine; Fort Orange/Albany and New Amsterdam (the Dutch trade); Fort Astoria present, Fort Okanogan, Fort Nez Percés present, Spokane House present, Fort Vancouver present, Fort Nisqually, Fort Langley present, Fort Victoria present, Fort St. James present, Fort McLeod present, Fort Fraser, Fort George (BC) present; Sitka/New Archangel (Russian), Yuquot present, Fort San Miguel present; Fort Enterprise, Fort Confidence, Fort Hope and Repulse Bay (Rae), Beechey Island (a discovery row exists), Fort Ross (Bellot Strait); Churchill: Prince of Wales Fort present; Port Nelson and the first York Factory; Henley

House, Albany present, Moose Factory present, Eastmain House, Rupert House present; Port-Royal present (on the wrong waterway), Louisbourg (event exists), Halifax (event exists), Red Bay present, L'Anse aux Meadows (only as a discovery; the field trip has no pin to open).

For portages (the one location type the legend filters that is not a building): Methye, Grand, Chaudière, Long Sault, La Vase, French, Frog, Echimamish, The Pas, Rat, Slave Falls, Great Dog and Hauteur des Terres are present; absent are the Mattawa portages, Portage la Prairie (a pronunciation term exists), Pigeon River's Grand Portage is present, the Athabasca Pass and Howse Pass (Thompson), the Yellowhead, the Chicago portage, the Fox-Wisconsin portage, and the Mohawk-Hudson carries.

5.3 People

The app has two person tables and one rule gap (2.3, 2.9). Candidates are listed by role; each needs a decision on table membership under the rule the owner chooses.

Explorers, traders and officers absent. Jolliet (present) and Marquette (present) have no waterway; Jean Nicolet has Lake Michigan only. Absent: Louis Hennepin; Daniel Greysolon Dulhut (Lake Superior, Lake Nipigon); Pierre de Troyes and the 1686 James Bay raid; Henry Kelsey is present; James Knight present; Jens Munk (Churchill 1619); Thomas Button present; Christopher Middleton and William Moor (Wager Bay 1742, 1746); Joseph Frobisher and Thomas Frobisher (Churchill River posts, the Montreal partners); Alexander Henry the Elder present; Peter Pond present; the La Vérendrye sons present without waterways; Jacques Repentigny Legardeur de Saint-Pierre; Anthony Henday present; Matthew Cocking present without a waterway; William Tomison present without a waterway; Joseph Colen likewise; John Rowand (Fort Edmonton); Colin Robertson (Athabasca campaign); John Stuart (New Caledonia, Fort St. James); Peter Skene Ogden (Snake Country, Fort St. James, the Pacific department); Samuel Black (Finlay River); Daniel Harmon present without a waterway; John McLoughlin present without a waterway (Fort Vancouver is a location); John Jacob Astor and the Astorians Gabriel Franchère, Alexander Ross, Duncan McDougall, David Stuart (the Pacific Fur Company already appears in the Columbia fur-trade row); Lord Selkirk, Miles Macdonell, Robert Semple (Red River colony, Seven Oaks); George Simpson present without a waterway; Thomas Simpson and Peter Warren Dease (Arctic coast 1837-39); John Richardson (named by three notable figures, absent as a person); John Ross and James Clark Ross (magnetic pole 1831); Francis Leopold McClintock (Fox 1859); Jane Franklin; Elisha Kent Kane and Charles Francis Hall (American Franklin searches, if the Arctic story is told fully); Frederick Schwatka; Jerry Potts; Pierre Falcon (the Métis bard); Alexander Ross; James Sinclair; Jaco Finlay (Howse Pass, Spokane House); David Thompson present; Simon Fraser present; John Stuart; George Vancouver present; Peter Puget and Joseph Whidbey if Puget Sound is added; Alexander Baranov and Grigory Shelikhov (Russian-American Company); Juan José Pérez Hernández present as Juan Pérez; Esteban José Martínez (Nootka Crisis 1789); Manuel Quimper (Juan de Fuca 1790); Francisco de Eliza; John Meares (Nootka); Charles Barkley and Frances Barkley (Juan de Fuca 1787, she the first European woman on the coast); Robert Gray present; Meriwether Lewis and William Clark (named by Sacagawea's row, absent as people); Zebulon Pike (Mississippi headwaters 1805) only if the Mississippi is added; Henry Youle Hind and John Palliser (present, no waterway) with Thomas Blakiston and James Hector; Robert Campbell (Yukon, Fort Selkirk) if the Yukon is to have an explorer; John Bell and Alexander Hunter Murray (Fort Yukon is present with no explorer).

Cartographers absent (the table has no surface today, so these matter only with a screen or with location links): Jean-Baptiste-Louis Franquelin (New France), Guillaume Delisle, Nicolas Sanson, Vincenzo Coronelli, Pierre Desceliers (Dieppe school, the Cartier coast), John Mitchell (1755), Thomas Jefferys, Henry Popple, Lewis Evans, Peter Pond as cartographer (his 1785 map is already a document), Aaron Arrowsmith present, James Cook and George Vancouver as surveyors, Joseph Bouchette present, Henry Youle Hind, Palliser's map, Franklin's and Back's survey maps, Rae's. **Indigenous cartographers**, central to the "cartography" theme and absent entirely: Ac ko mok ki (Old Swan), whose 1801 map for Peter Fidler guided Lewis and Clark and Arrowsmith; Ki oo cus; Cha chay pay way ti (Cree, for Fidler);

the Chipewyan maps Matonabbee and Idotliazee drew for Moses Norton that set Hearne's course; Iligliuk's maps for Parry and Lyon (1822); Tookoolito and Ebierbing with Hall; the Inuit maps Rae and McClintock relied on. These link directly to Fidler, Norton, Hearne, Parry and Matonabbee, all present.

Indigenous leaders, guides and interpreters absent (surface: Notable Figures; link by exact waterway name): Maquinna and Wickaninnish (Nootka Sound, Clayoquot; Cook, Meares, Martínez, Vancouver); Kwah (Carrier, Fort St. James, Simon Fraser, John Stuart); Membertou (Port-Royal, Champlain); Donnacona, Domagaya and Taignoagny (Cartier, Stadacona); Kondiaronk (Great Peace of 1701); Tessouat (Allumette Island, Champlain 1613); Joseph Brant (the Haudenosaunee and the Grand River); Pontiac and Tecumseh are later and marginal to waterways but tied to Detroit; Maskepetoon; Piapot; Mistawasis and Ahtahkakoop (Treaty 6 at Fort Carlton, a field trip exists); Peguis is present as an explorer row only; Yellow Quill; Nescambiouit; William Ouligbuck and his father (Inuit interpreters for Franklin, Rae); Mikak (Labrador Inuit, London 1768); Tookoolito and Ebierbing; Demasduit and Nonosabasut (Beothuk, with Shawnadithit present); English Chief (Mackenzie 1789); Awgeenah; Nestabeck (Keskarrah is present); Augustus/Tattannoeuck present and Junius/Hoeootoerock absent; Saukamappee present; Old Swan as cartographer above; the Mandan chiefs Sheheke and Black Cat (Thompson 1797, Lewis and Clark); Comcomly (Chinook, Astoria, Fort Vancouver); Concomly's daughter Ilchee; the Tsleil-Waututh and Squamish leaders Vancouver met (Burrard Inlet is present).

Métis figures absent. Pierre Falcon; Jean-Baptiste Lagimodière present; Marie-Anne Gaboury present; Cuthbert Grant present; James McKay (Treaty interpreter); Peter Erasmus (Palliser's guide, Treaty 6 interpreter); Jerry Potts; Louis Goulet; Gabriel Dumont present; Marie-Rose Delorme Smith; Pierre Bottineau (Red River guide to the US surveys); Johnny Grant.

Women absent. Frances Barkley (above); Amelia Douglas (Fort St. James, Fort Vancouver, Victoria); Letitia Hargrave (York Factory letters, a primary source); Frances Simpson; Jane Franklin; Marguerite Bourgeoys and Jeanne Mance only if Montreal's founding is told; Hélène Boullé (Champlain); Marie de l'Incarnation (letters on New France); Thanadelthur, Charlotte Small, Isabel Gunn, Marie-Anne Gaboury, Sacagawea present; the "country wives" named in Thompson's and Harmon's journals; Catherine Sinclair; Nahovway; Mary Sinclair Inkster.

5.4 Teaching resources, by gap

- **Quizzes.** No quiz on New France, the Northwest Passage, the Pacific voyages, cartography, Métis history, women, treaties or Indigenous leaders; one quiz has one question; six are duplicates; K-3 and 10-12 each have one distinct quiz. The category list is hard-coded to five, so a "Cartography" or "Métis History" quiz would be unfilterable without a client change.
- **Field trips.** None for K-3 or 10-12; none on the St. Lawrence, the Pacific, the Arctic, the Great Lakes, or any Indigenous site except as the Norse met the "Skrælings". Each new trip should set `locationId` on its stops once a screen reads them.
- **Documents.** 34 without a grade; none for K-3 or 10-12; nothing from Marquette and Jolliet (Marquette's 1673 journal), Hennepin, La Vérendrye's own journals, Hearne's "A Journey" is represented, Thompson's "Narrative" by two; nothing by Franklin, Richardson, Parry, Rae's own letters, Harmon's journal, Henry the Younger's journal, Franchère, Ross, Cook's and Vancouver's journals, the Spanish Nootka accounts, Letitia Hargrave's letters, Isabel Gunn's record, the HBC post journals as a type; one image across all 48.
- **Timeline and What Happened Here.** The timeline stops at 1885 and starts at 1613; no Norse, Cabot, Cartier, Champlain 1608, Hudson, Munk, Kelsey, Cook, Vancouver, Nootka Crisis, Astoria, Oregon Treaty 1846, Franklin, Rae, Amundsen. Historical events have the Cartier, Champlain, Cook, Franklin and Amundsen items but nothing on the Pacific Spanish, the Russians, the Dutch, the Mississippi, or any treaty before 1871.

- **Indigenous languages.** Ten language labels for a country of some seventy; no words in Halkomelem, Nuu-chah-nulth, Dakelh (Carrier), Tlingit, Innu, Wolastoqey, Michif (the Métis language), Anishinaabemowin as distinct from "Ojibwe", Nakoda, Tsuut'ina; three stories; stories and vocabulary quizzes cannot be opened; zero audio anywhere; zero word links to waterways although 27 words are in the "waterway" category.
- **Journeys.** Two. Natural additions with coordinates the app can already draw: the Methye Portage brigade, Hearne to the Coppermine, the Red River cart trail, Fraser's descent, Thompson to the Pacific 1811, a York boat from York Factory to Norway House, Champlain 1615 to Huronia, Jolliet and Marquette 1673.
- **Comparisons.** Four; none at K-3 or 4-6. The generated explorer and fort comparisons are built server-side and unused.
- **Printables.** None at K-3; no map of the Pacific, the Arctic or New France.

5.5 Imagery and geometry, what can be asserted

- Every current image except three Wikimedia URLs is on R2 (`pub-3768…r2.dev`). The last health run (2026-10-16 02:01) found 3 broken records: Marble Island gallery[0] (commons.wikimedia.org), the Crowfoot portrait used by both NotableFigure `crowfoot_leader` and HistoricalEvent `cmlhqohkg000gm2m4tpltfb4p`, and one rate-limited Coppermine image. The health job logged each three times in one run (6,150 rows total), which looks like three scheduler instances; a code item, not content.
- R2 credentials are not present in this session's environment (`R2_*` absent from `backend/.env`), so no upload can be verified here; `backend/src/lib/image-cdn.ts` (`uploadImageFromUrl`) is the path when they are.
- Geometry: `export-waterway-kml.ts` queries OpenStreetMap by name within a North American bbox and can take an explicit `--osm=relation/ID`. Whether a relation exists and is clean for any waterway in 5.1 is **unverified** until the exporter is run for it with the continuity guard; the St. Lawrence experience (centreline plus surface polygons) shows wide estuaries need `--with-surface`.

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6. Decisions the owner must make before any writes

1. **Who is an Explorer.** Either (a) keep "anyone a Deep Dive key figure points at is an Explorer row" and accept Riel, Crowfoot and Thanadelthur in the Explorers tab, or (b) add a `notableFigureId` alternative to `keyFigures[]` and `useNotableFigure` in the lesson screen, then move the fourteen to NotableFigure only. (b) is a code change plus a data move.
2. **Duplicates.** Nations ×7, Cartographers ×2, Quizzes ×6: delete the later-created copies after confirming no attempt or link references them (quiz attempts: 0 today).
3. **Fort pairs** (2.2): keep both as distinct posts with distinct years and notes, or merge.
4. **Taxonomies.** Canonical `locationType` (fold `fort`, `historic_site`), `WaterwayType` (fold `lake`), `Explorer.nationality` (controlled list, free text kept in biography), `FurTradeInfo.tradingCompany` spellings, `NotableFigure.figureType` colour config for `metis_leader` and `voyageur`, hard-coded quiz categories and lesson topics if new themes are added.
5. **Document grade levels** for 34 rows, and whether Documents should have a grade filter UI like every other resource list.
6. **Mackenzie River (Ontario)** and the Roderick McKenzie figure row that reaches it.
7. **Port-Royal's waterway** (St. Lawrence today) and the two Churchill-mouth forts on the Saskatchewan Churchill row.

8. **Assessment field name** in the teacher Lesson Plan screen (`assessmentSuggestions` vs `assessment`): fix the type and screen, or add an alias on the server.
9. **The twelve unreachable achievements**: record `waterway_explored` and `location_visited` from the detail sheet, `document_read` from the document screen, `pronunciation_learned` from the pronunciation screen, or retire them.
10. **Stories and vocabulary quizzes**: give them screens, or stop listing them.
11. **Links no screen reads** (mechanism C): decide which to make navigable before filling them, in this order of value: field trip stop → pin and waterway; timeline/What Happened Here → explorer, waterway, location; document → explorer; notable figure → explorer by id.
12. **Static reading guides**: fix the 4 dead keys and the escaped author strings, add the 2 waterways and 17 locations, and accept that each new place or person needs a build, or move the guides to Turso.

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7. What this sweep did not verify

- The historical accuracy of existing prose (descriptions, biographies, notes). Only structure, links and counts were checked.
- The content of `videoUrl` JSON beyond its shape (69 waterways, 100 locations carry a value; the health job does not check video URLs).
- OpenStreetMap coverage for any waterway in 5.1; portrait provenance for any person in 5.3.
- `backend/src/routes/teachers.ts`, `auth.ts`, `admin-approval.ts`, `teacher-approval.ts`, `sample.ts` were read earlier in the project but are account plumbing and carry no content.

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8. Next steps (nothing below starts without approval)

1. Owner reads this document and answers section 6.
2. Fix-first data pass under the established pattern: dry run, committed backup under `backend/prisma/turso-migrations/backups/`, `--apply`, verification, idempotent rerun.
3. Code items that gate content value: lesson key-figure link to notable figures (if 6.1b), stop/timeline/document link reading, achievement recording, story and vocabulary quiz screens, grade filter on Documents, assessment field, taxonomy configs, reading guides.
4. Content packs by region and theme, student and teacher faces together, each pack validated against section 2 (required fields, canonical values, link targets that exist, exact-name strings for text matches) and section 5 (relevance stated), images fetched with credit to R2, KML exported with the guard and the overview rebuilt for rivers, then the same dry run → backup → apply → verify cycle.
5. Sources for authoring: Dictionary of Canadian Biography, The Canadian Encyclopedia, Library and Archives Canada, Hudson's Bay Company Archives (Archives of Manitoba), Champlain Society editions, Wikimedia Commons for imagery with licence recorded, OpenStreetMap for geometry.

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9. Decisions taken and implementation status (2026-10-16, later the same day)

The owner answered all twelve questions in section 6: 1 (b); 2, 4, 11 as recommended; 3 merge; 5 yes with a grade filter; 6 yes, provided the real Mackenzie River's display is untouched; 7 yes; 8 fix the type and screen, no alias; 9 record, do not retire; 10 build the screens; 12 move the guides to Turso and identify grade-level gaps.

9.1 Code phase (done, this branch)

- Key figures may carry `notableFigureId` as well as `explorerId`; the Deep Dive screen opens the Notable Figure screen for the former and takes the portrait from whichever record is linked (`lessons/[id].tsx`, `types/education.ts`).
- Documents list has a grade chip row; the server already filtered on `gradeLevel`.
- `assessmentSuggestions` renamed to `assessment` in the type and both teacher screens.
- Colour and label config added for `metis_leader`, `voyageur`, live pronunciation term types, document types `report` and `inscription`, comparison types `perspectives`, `trade_companies`, `trade_routes`, and the Blackfoot, Mohawk, Iroquois and Inuit labels. Underscore replacement is now global in the four screens that only replaced the first one.
- Exploration points: new table `UserExploration` and route `POST /api/gamification/explore`; the server fixes the points (waterway 5, location 3, document 5, pronunciation 2) and awards each (device, activity, record) once. Opening the detail sheet records waterway or location, opening a document records it, and the pronunciation card gained an "I've learned this" button. The twelve achievements of section 2.17 are reachable.
- Screens `stories/[id]` and `vocabulary-quiz/[id]`; the cards on the Indigenous Languages screen open them.
- Related records are navigable: the server attaches `explorer`, `waterway`, `location` objects to timeline events, What Happened Here events, documents and field-trip stops (`backend/src/lib/related.ts`), notable-figure associated explorers gain a resolved `explorerId` (exact or honorific-stripped name match), and the map accepts `/?markerId=&markerType=` to open a record and `/?lat=&lng=` to centre. A shared `RelatedLinks` component renders the chips.
- Reading guides live in the new `ReadingGuide` table and are returned inside the waterway, location, explorer and notable-figure detail responses; the two static files are deleted.

9.2 Reading-guide import and the grade-level gap

`2026-10-16-import-reading-guides.ts` moved the static entries into Turso (backup `backups/2026-10-16-reading-guides-import.json`).

Measure	Value
Entries in the static files	1,047
Exact duplicates (same title, source and url) skipped	60
Rows inserted	986 (waterway 254, location 345, explorer 387)
Targets covered	68 waterways, 85 locations, 92 explorers
Fields truncated at an apostrophe in the static files	290
Of those restored to the full text (unambiguous titles and publishers)	133
Of those kept as the fragment before the cut	157

Measure	Value
Entries dropped	1 (Batoche: Diane Payment, Golden Dog Press, title lost entirely)
Four dead keys remapped by the place their entries describe	Fort Simpson, Oxford House, Fort Edmonton (HBC), Rocky Mountain House (NWC)

Grade bands. No entry carried a reading level. Each row was given one by source type, a heuristic the owner can override per row: encyclopedias, Parks Canada, museums and Wikipedia pages read as 7-9; archival series, scholarly editions and monographs as 10-12.

Band	Entries	Targets with at least one entry
K-3	0	0 of 262
4-6	0	0 of 262
7-9	437	all 245 covered targets have at least one encyclopedia-level entry
10-12	549	all 245 covered targets

So the reading-guide gap is uniform: there is no K-3 or 4-6 "Additional Study" material for any waterway, location or explorer, and no guide at all for Hayes River, Kaministiquia River and fifteen locations (Lower Fort Garry, Fort Ellice, Fort Kaministiquia, Sault Ste. Marie Post, Fort Augustus (NWC), Fort George (NWC - Saskatchewan), Fort McLeod (BC), Fort George (NWC - BC), Fort Gibraltar, Fort Lac La Pluie, Fort Battleford, Fort Liard, Fort Simpson (BC), Fort McLoughlin, Rupert House). Notable figures have none yet; the fourteen people moving out of `Explorer` bring their guides with them in the data pass. Filling the K-3 and 4-6 bands means children's and middle-grade titles (picture books, Canadian Geographic Kids, Scholastic Canada biographies, Parks Canada learning pages), one or two per target, sourced at authoring time.

9.3 Data phase (done, same day; one script per decision under `backend/prisma/turso-migrations/`, each with a dry run, a committed backup and a verification step)

Decision	Script	Result (verified by query after apply)
2 duplicates	`2026-10-16-dedupe-nations-cartographers-quizzes.ts`	7 nations, 9 cartographers, 7 quizzes with 36 questions; the one class assignment that pointed at a deleted quiz copy repointed to the kept copy
6 bogus river	`2026-10-16-remove-mackenzie-ontario.ts`	row and its 4 guide entries deleted; Roderick McKenzie now names Fort William, Thunder Bay, Lake Superior; the real Mackenzie River untouched
1b people	`2026-10-16-move-people-to-notable-figures.ts`	15 Explorer rows removed (77 remain); figure rows created for Chief Peguis and Shawnadithit (20 figures); 35 key-figure entries in 12 lessons now carry `notableFigureId`; 64 reading-guide rows retargeted; portraits copied to Akaitcho, Chief Peguis, Cuthbert Grant, Gabriel Dumont, Lagimodière, Riel, Gaboury, Shawnadithit. **Not copied:** Thanadelthur (no likeness exists). **Provenance to confirm:** Cuthbert Grant, Jean-Baptiste Lagimodière, and the pre-existing Matonabbee portrait; none has an authenticated contemporary likeness that the author of this document knows of

Decision	Script	Result (verified by query after apply)
4 taxonomies	`2026-10-16-canonical-taxonomies.ts`	locationType: Fort 53, Trading Post 16, Portage 14, Cultural Site 12, Settlement 5; WaterwayType `lake` folded (Lake 13); nationality reduced to 16 values with 11 biographies gaining one sentence of detail; tradingCompany to 10 full-name spellings
3 fort pairs	`2026-10-16-merge-fort-pairs.ts`	94 locations; Fort William, Fort Edmonton, Fort Churchill, Upper Fort Garry (absorbing Fort Garry and Fort Garry (Upper)), Rocky Mountain House each hold the merged notes, gallery, earliest year and reading guide; Riel's associated location "Fort Garry" rewritten
7 waterways	`2026-10-16-bay-of-fundy-and-churchill.ts`	see below
5 grades	`2026-10-16-document-grades.ts`	applied 2026-10-17 after approval: no document without a band; 4-6 ×10, 7-9 ×28, 10-12 ×10

Bay of Fundy. New waterway (type Bay, anchor 45.1 N 65.6 W, region New Brunswick / Nova Scotia). Geometry from OpenStreetMap through the exporter's bay branch: one relation, 15,148 points, bbox 66.95 W to 64.94 W, 44.21 N to 45.49 N, anchor inside. Four Wikimedia Commons images with licence and author recorded (Eric Van Lochem CC BY-SA 4.0; Ian Mackenzie CC BY 2.0; Madereugeneandrew CC BY-SA 3.0; NASA/METI ASTER public domain), served as `Special:FilePath` URLs because the R2 keys are not in this session. Champlain linked (1604) with expedition notes; three reading-guide entries; Port-Royal moved onto it. The Mi'kmaq name is left null rather than asserted.

Churchill River: a correction to decision 7. The row named "Churchill River (Saskatchewan)" already carried the whole river: its KML spans 108.38 W to 94.18 W and reaches 58.79 N, the mouth at Churchill, Manitoba. A separate "Manitoba" row would have duplicated that geometry, so the row was renamed "Churchill River" (region unchanged: Saskatchewan / Manitoba) and the merged Fort Churchill moved onto it from Hudson Bay, beside Prince of Wales Fort, Methye Portage and Frog Portage. The home-map overview was rebuilt for the new name (`generated/river-overview.json`, 28 rivers).

9.5 Content pack 1, New France to the Gulf (applied 2026-10-17)

Data in `backend/prisma/turso-migrations/packs/new-france-to-the-gulf.ts`, applied by `2026-10-17-pack-new-france-to-the-gulf.ts` after a dry run that resolved every name to a live id and confirmed every image URL and KML file; backup of the backfilled events in `backups/`. Idempotent: a rerun reports everything as existing and writes nothing.

Added	Count	Notes
Waterways	11	Mississippi, Illinois, Wisconsin, Fox (Wisconsin), Richelieu, Lake Champlain, Hudson, Lake Nipissing, Mattawa, Detroit, Niagara; all with OpenStreetMap geometry (the Fox re-exported with a 25 km continuity gap so the Illinois Fox was not chained to it); Indigenous names only where the attestation is well known, language stated; fur-trade rows for the Mississippi and the Hudson
Explorers	4	Hennepin, Dulhut, de Troyes, Tonti; no portraits (no verified likeness)

Added	Count	Notes
Explorer–waterway links	37	Jolliet, Marquette, La Salle, Cadillac, Champlain, Hudson, Nicolet, Brûlé and Iberville now have waterways; Abitibi, Moose and Kaministiquia rivers gain their first explorer
Locations	21	from Stadacona and Hochelaga to Fort Orange, New Amsterdam, Michilimackinac, Fort de Chartres, Kaskaskia and New Orleans; Fort Crèvecoeur, Stadacona and Hochelaga have no image (nothing suitable and licensed on Commons)
Notable figures	4	Donnacona, Kondiaronk, Tessouat, Membertou; associated locations use exact waterway names so they appear on those waterways
Timeline events	17	1535 to 1763, each linked to its explorer, waterway or location
What Happened Here events	9 new, 14 existing events given their explorer/waterway/location links	
Pronunciations	17	
Quiz	1	8 questions, grade 7-9, every question with `sourceType` / `sourceId`
Deep Dive / lesson plan	1	grade 7-9, student and teacher faces, six key figures (five explorers, Kondiaronk as a figure), seven timeline items linked to events, related waterway/location/explorer/quiz ids filled
Field trip	1	the first K-3 trip, Michilimackinac, five stops each linked to the pin and Lake Huron
Reading-guide entries	69	Wikipedia and Canadian Encyclopedia at 7-9, Dictionary of Canadian Biography, Jesuit Relations and Parkman at 10-12
Images	63 URLs	all Wikimedia Commons, public domain or CC BY / CC BY-SA, author and licence in every credit, served as `Special:FilePath` URLs until R2 keys are available

Added later the same day (`2026-10-17-primary-sources-new-france.ts`): 17 primary-source pointer documents, each saying what the source is, who made it and when, and exactly where it is readable today (archive, edition, verified URL), with no quoted transcription; linked to their explorer, waterway and location and graded (1 at 4-6, 10 at 7-9, 6 at 10-12). Still not included: Indigenous vocabulary. The river overview was rebuilt (37 rivers). The Fox River trip and the `FieldTripStop.narrative` field: the trip table has no introduction or conclusion columns, so that text stays in the pack file.

9.6 Content pack 2, The Pacific and the Russians (applied 2026-10-17)

Data in `backend/prisma/turso-migrations/packs/pacific-and-russians.ts`, applied by `2026-10-17-pack-pacific-and-russians.ts` (dry run, backup of every row it changed, apply, verify; idempotent). Scope: the Northwest Coast from Bering (1741) to the Oregon Treaty (1846) and the sale of Alaska (1867): Spain and the Nootka Crisis, Cook and Vancouver, the Russian-American Company at Sitka, Astoria, the North West and Hudson's Bay companies west of the Rockies, Lewis and Clark, and the Indigenous leaders who dealt with all of them.

Added	Count	Notes
Waterways	14	Puget Sound, Prince William Sound, Cook Inlet, Sitka Sound, Dixon Entrance, Clayoquot Sound, Snake, Willamette, Kootenay, Thompson, Skeena, Stikine, Nass, Okanagan Lake. OpenStreetMap geometry for ten; the four Alaskan bodies (Prince William Sound, Cook Inlet, Sitka Sound, Dixon Entrance) are mapped in OSM as single nodes with no shoreline polygon and so are pins only. The existing Georgia Strait row, which had no geometry, was given its OSM polygon (719 KB). Fur-trade row for the Snake
Explorers	14	Ogden, John Stuart, Samuel Black, Martínez, Quimper, Meares, C. W. Barkley, Puget, Alexander Ross, Franchère, David Stuart, Lewis, Clark, Jaco Finlay. One portrait: Meares, from W. Beechey's engraving in the 1790 *Voyages* (a lifetime likeness, confirmed from the Commons file's own record). No usable lifetime likeness was found on Commons for the others, so they have none
Explorer–waterway links	44	Cook, Bering, Pérez, Vancouver, Gray, Quadra, Fraser, Thompson, McLoughlin, Douglas, Simpson and Harmon gain the new waterways; every new explorer has at least one
Locations	11	Fort Okanogan, Fort Nisqually, Novo-Arkhangelsk (Sitka), Fort Fraser, Fort Boise, Fort Clatsop, Celilo Falls, Opitsaht, Fort Stikine, Howse Pass, Athabasca Pass. Fort Kamloops moved from the Fraser River row to the new Thompson River (it stands where the North and South Thompson meet; backup records the old waterway id)
Notable figures	9	Maquinna, Wickaninnish, Comcomly, Kwah, Amelia Douglas, Frances Barkley, John Jacob Astor, Alexander Baranov, Grigory Shelikhov. Two portraits: Astor (Gilbert Stuart, 1794) and Lady Douglas (a photograph taken in her lifetime, NARA). Maquinna's biography notes that the Dictionary of Canadian Biography treats the chief of 1778-1795 and the chief who seized the *Boston* in 1803 as two men
Timeline events	28	1741 to 1867
What Happened Here events	13 new; 3 existing Pacific events (Vancouver meets Galiano and Valdés, Fort Victoria established, Juan de Fuca Strait discovered) given their explorer/waterway/location links	
Pronunciations	17	Nuu-chah-nulth, Chinook, Dakelh, Secwepemctsín, Nsyilxcən, Ktunaxa, Tlingit, Dena'ina, Haida and Russian terms
Quiz	1	"The Pacific Contest", 8 questions, grade 7-9, every question sourced

Added	Count	Notes
Deep Dive / lesson plan	1	"The Pacific Contest: Four Empires on the Northwest Coast, 1741-1846", grade 10-12, nine key figures (five explorers, four notable figures incl. Maquinna, Baranov, Astor and Amelia Douglas), ten timeline items linked to events
Field trip	1	"Yuquot and the Nootka Crisis of 1789", grade 10-12, six stops at Friendly Cove and Nootka Sound
Reading-guide entries	73	Wikipedia and Canadian Encyclopedia at 7-9, Dictionary of Canadian Biography and the Lewis and Clark journals site at 10-12
Images	75 URLs	Wikimedia Commons, public domain or CC BY / CC BY-SA, author and licence in every credit; includes Webber's 1778 view of the houses at Nootka, the 1790 print of the Argonaut's seizure, Postels's 1827 view of Castle Hill, William Hind's Athabasca Pass (c. 1862), Clark's map of the Great Falls of the Columbia, Franchère's drawing of Astoria in 1813, and the 1911 Admiralty chart of Dixon Entrance

Two corrections made along the way. Wikimedia's API throttled this session's shared address on every endpoint, so the image search was rebuilt to use the ordinary HTML search page and each file's raw wikitext (same manifest shape; copy in `backups/2026-10-17-commons-pack2-manifest.json`). And biographi.ca answers 200 with a "Biography not found" page for any slug, so a status check proves nothing: the pack script now checks every DCB guide URL by page title, nine guessed slugs in the pack were corrected or replaced before applying, and an audit of the DCB guide URLs already in Turso found and fixed two wrong ones (Tonty, Tessouat; `2026-10-17-fix-dcb-guide-urls.ts`).

Added the same day (`2026-10-17-primary-sources-pacific.ts`): 23 primary-source pointer documents (Cook's Nootka journal, Meares's Memorial and Voyages, the Nootka Convention and the 1790 official papers, Mozino, Vancouver, Jewitt, the Lewis and Clark journals, Franchère, Ross, Cox, Thompson's Narrative, Harmon, Lisiansky on the Battle of Sitka, Steller, Simpson's 1824-25 journal, Ogden's Snake Country journals, McLoughlin's letters, Frances Barkley's reminiscences, Pérez's 1774 diary, Irving's Astoria, the Oregon Treaty, the Alaska cession treaty), each with what it is, who made it and when, and where it can be read (archive, edition, URL only where the page answered), linked and graded (12 at 7-9, 11 at 10-12). Documents now total 88 (11 at 4-6, 50 at 7-9, 27 at 10-12). The river overview was rebuilt (44 rivers). Data only: no mobile build is needed for this pack.

9.7 Content pack 3, The Arctic Searches (applied 2026-10-17)

Data in `backend/prisma/turso-migrations/packs/arctic-searches.ts`, applied by `2026-10-17-pack-arctic-searches.ts` (dry run, backup, apply, verify; idempotent). Scope: the Northwest Passage from Frobisher (1576) to Amundsen (1906) and Bernier (1909), approached from the Atlantic (Davis Strait, Hudson Strait, Lancaster Sound) and from the Pacific (Bering Strait, the Beaufort Sea), the Franklin search of 1848-1859 as the decade that mapped the archipelago, and the Inuit who guided, interpreted for and finally explained the fate of the expeditions.

Added	Count	Notes
Waterways	16	Hudson Strait, Beaufort Sea, Bering Strait, Coronation Gulf, Victoria Strait, Peel Sound, Prince Regent Inlet, Wellington Channel, Viscount Melville Sound, Prince of Wales Strait, Smith Sound, Simpson Strait, Amundsen Gulf, Repulse Bay (Naujaat), Wager Bay (Ukkusiksalik), Kotzebue Sound. OpenStreetMap polygons for fifteen; Kotzebue Sound, and the existing Davis Strait and Parry Channel rows, are single OSM nodes and stay pins. The Hudson Strait polygon (331,000 points, 8.6 MB) was thinned to every fifth point (2.2 MB); the full export is kept outside the repository. Seas and gulfs use the Bay type, as Queen Maud Gulf already did
Explorers	20	John Ross, James Clark Ross, Richardson, Thomas Simpson, Dease, McClintock, Collinson, Beechey, Kotzebue, Kane, Hall, William Kennedy, Bellot, Crozier, Middleton, Munk, Sverdrup, Bernier, Penny, Kellett. Portraits for eight, all lifetime likenesses confirmed from the Commons file records: J. C. Ross (1852 print), Richardson, McClintock, Collinson and Beechey (Stephen Pearce, 1850s), Crozier (Beard daguerreotype, 1845), Hall (the only known photograph), Bernier (Bain photograph)
Explorer–waterway links	100	Frobisher, Hudson, Button, Bylot, Baffin, Foxe, James, Cook, Parry, Franklin, Back, Rae, McClure, Amundsen, Stefansson, Hearne and Mackenzie gain Arctic waterways; Button, Bylot, James and Amundsen had none before
Locations	15	Beechey Island, Fort Enterprise, Fort Franklin (Délı̨nę), Fort Confidence, Winter Harbour, Victory Point, Gjoa Haven (Uqsuqtuuq), Fort Hope, Kodlunarn Island, Mercy Bay, Felix Harbour, Herschel Island (Qikiqtaruk), Icy Cape, Port Leopold, Dealy Island. Fort Confidence and Victory Point have no image (nothing suitable and licensed on Commons)
Notable figures	11	Kalicho, Arnaq, Ouligbuck, William Ouligbuck, In-nook-poo-zhe-jook, Tookoolito, Ebierbing, Eenoolooapik, Lady Jane Franklin, Hans Hendrik, Qillarsuaq. Portraits for four: John White's 1577 watercolours of Kalicho and of Arnaq with Nutaaq, the tintype of Tookoolito and Ebierbing, the Smithsonian photograph of Ebierbing
Timeline events	36	1577 to 1913; the two unlinked Mackenzie timeline events were given their explorer and waterway (new `timelineBackfill` in the pack format)
What Happened Here events	17 new; 11 existing events linked (Franklin departs, ships trapped, Erebus and Terror found, Amundsen, Hearne's three expeditions, surrender of Prince of Wales Fort, creation of Nunavut, Bloody Falls ceremony)	
Pronunciations	18	Inuktitut, Inuinnaqtun and Inuvialuktun terms
Quiz	1	"The Search for the Northwest Passage", 8 questions, grade 7-9

Added	Count	Notes
Deep Dive / lesson plan	1	"Two Oceans, One Passage: The Arctic Searches, 1576-1909", grade 10-12; twelve key figures (seven explorers, five notable figures); the existing 7-9 lesson "The Northwest Passage: Dreams and Disasters" is untouched
Field trip	1	"Beechey Island: Franklin's First Winter", grade 7-9, six stops
Reading-guide entries	112	every Dictionary of Canadian Biography and Canadian Encyclopedia link verified by page content, every Wikipedia link by HEAD, before applying
Images	74 URLs	Commons, public domain or CC BY / CC BY-SA, credited; includes John White's 1577 skirmish at Bloody Point, Parry's 1826 view of Prince Regent Inlet, the 1862 Admiralty chart of Wellington Channel and Melville Island, McDougall's watercolour of the Investigator in Mercy Bay, the Fox, Hecla and Griper at Winter Harbour, the Beechey Island graves, and the Library of Congress chart of the coast from the Mackenzie to Bering Strait

Link repair found on the way: with a browser user agent the Canadian Encyclopedia can be checked after all (404 for a missing article), and 48 of the 164 Canadian Encyclopedia guide links already in Turso were dead, guessed slugs from the October 16 import. `2026-10-17-fix-tce-guide-urls.ts` replaced 47 of them with verified Canadian Encyclopedia, Dictionary of Canadian Biography or Wikipedia pages (each candidate tested live at dry run) and deleted the one with no working replacement; all three pack scripts now check those two hosts by content. Reading guides total 1,117 after the repair.

Added the same day (`2026-10-17-primary-sources-arctic.ts`): 26 primary-source pointer documents, from Best's True Discourse (1578), Davis, Prickett on Hudson, Baffin, Munk and Middleton through Parry, Franklin's two Narratives, Beechey, Kotzebue, John Ross, Simpson, Rae, the Victory Point record, McClintock, McClure and Miertsching, Kennedy and Bellot, Kane and Hans Hendrik, Hall's two books and the Inuit testimony, Richardson, Amundsen, Bernier, Sverdrup and Stefansson; each with what it is, who made it and when, and where it can be read (archive, edition, URL only where the page answered), linked and graded (12 at 7-9, 14 at 10-12). Documents now total 114 (11 at 4-6, 62 at 7-9, 41 at 10-12). Data only: no mobile build is needed.

9.8 Gaps reported by the owner on 2026-10-17, and the first repairs

The owner reported, on the production app: no Slave River, North Saskatchewan or South Saskatchewan; the Mattawa River narrative names Champlain and Brûlé but the explorer panel does not show them; many explorers without portraits where likenesses exist (Lewis, Clark); and a general concern that the content leans on single sources. Findings and actions:

- **Narrative links.** A database-wide read of every waterway's text for explorer names with no link, and of every explorer's text for waterway names with no link, found 36 real gaps (after discarding false hits such as "Hudson's Bay Company" for Henry Hudson). All 36 links were written with a year and a sentence checked against the Dictionary of Canadian Biography or the explorer's narrative (`2026-10-17-link-gaps.ts`): Brûlé on the Mattawa, French River and Lake Nipissing; Button, Foxe and James on Hudson Bay; Groseilliers on Lake Superior and Hudson Bay; the Hudson's Bay Company's inland men (Kelsey, Henday, Hearne, Cocking, Turnor, Tomison) on the bay and the Saskatchewan; Roderick Mackenzie on Lake Athabasca; Galiano and Valdés on Burrard Inlet; and others. One wrong sentence was corrected with a backup: Chesterfield Inlet was explored in 1762 by Moses Norton, not by Hearne in 1770. The Mattawa River on production now returns Champlain (1615) and Brûlé (1610);

Champlain was already linked in the data before this, so if the panel showed neither on the device, the app was showing cached data.

- **Portraits.** Every explorer and notable figure without an image was checked against the lead image of its Wikipedia article and the Commons record of that file, and a Commons search was run for the rest. Twelve lifetime likenesses were confirmed and set (`2026-10-17-portraits-from-commons.ts`): Lewis (Peale, c. 1807) and Clark (Peale, 1810), as the owner pointed out; Hennepin (1694 engraving); Ogden (photograph before 1854); Martínez (Spanish naval portrait, c. 1785); Quimper (c. 1790); Alexander Ross (photograph); Franchère (before 1863); Sverdrup (photograph, National Library of Norway); Tattannoeuck (Robert Hood, 1821); Maquinna (Suría, 1791); Baranov (Tikhanov, 1818). Not set, because no likeness made in the person's lifetime exists or could be found: Sacagawea (every image is a twentieth-century statue or painting; the owner may decide whether a labelled depiction is acceptable), Thompson, Thanadelthur, Charlotte Small, Isabel Gunn, Saukamappee, Keskarrah (Hood's 1821 drawing of him exists at Library and Archives Canada but is not on Commons), Donnacona, Kondiaronk, Tessouat (the Codex canadensis "one-eyed captain" is c. 1700, posthumous), Membertou, Wickaninnish, Kwah, Comcomly (Catlin's portrait is posthumous), John Ross (the BnF image is a later reproduction), Kane (Hicks's portrait is 1858, after his death), Shelikhov and Kotzebue (undated), Bellot, Kennedy, Dease, Middleton, Munk, Penny, Kellett, Eenoolooapik, Lady Franklin, Hans Hendrik, and the eighteenth-century Hudson's Bay Company men.
- **Missing waterways.** The roster was compared with a list of the fur-trade and exploration rivers and lakes; 33 exports from OpenStreetMap are running for a fourth pack (Slave, North and South Saskatchewan, Liard, Winnipeg River, Rainy River and Lake, Sturgeon-Weir, Clearwater, Bow, Qu'Appelle, Finlay, Parsnip, Nechako, Stuart Lake, Bella Coola, Georgian Bay, Lake Timiskaming, Lakes Winnipegosis and Manitoba, Pigeon, Dease, Pelly, Kazan, Dubawnt, Seal, Severn, Rupert, Eastmain, Ungava Bay, Lakes Mistassini and Abitibi, Gulf of St. Lawrence).
- **Sources.** 327 records carry reading guides; 47 of them point to a single kind of source and 22 have none (Hayes and Kaministiquia rivers, fourteen posts, and the figures Sacagawea, Saukamappee, Tattannoeuck, Keskarrah and Roderick McKenzie). Overall the guides cite 348 Wikipedia pages, 129 Canadian Encyclopedia articles, 89 Hudson's Bay Company Archives references, 41 Parks Canada pages, 35 Dictionary of Canadian Biography entries and some 300 books and editions. The next pass adds a scholarly or archival source to every single-source record and guides to the 22 without.

9.9 Content pack 4, The Interior Rivers (applied 2026-10-17)

Data in `backend/prisma/turso-migrations/packs/interior-rivers.ts`, applied by `2026-10-17-pack-interior-rivers.ts` (dry run, backup, apply, verify; idempotent). Scope, from the owner's report of the missing Slave River and Saskatchewan branches: the water highways of the interior fur trade from Georgian Bay and the Pigeon River to the two branches of the Saskatchewan, the Sturgeon-Weir and Clearwater links to the Athabasca, the Slave and the Liard, and the New Caledonia rivers, told as Indigenous routes the traders followed.

Added	Count	Notes
Waterways	21	Slave, North Saskatchewan, South Saskatchewan, Liard, Winnipeg River, Rainy River, Rainy Lake, Sturgeon-Weir, Clearwater, Bow, Qu'Appelle, Finlay, Parsnip, Nechako, Stuart Lake, Bella Coola, Georgian Bay, Lake Timiskaming, Lake Winnipegosis, Lake Manitoba, Pigeon River. OpenStreetMap polygons or centrelines for twenty (Rainy Lake thinned from 149,000 to 50,000 points); Georgian Bay is a single OSM node and stays a pin. Fur-trade rows for the Slave, North Saskatchewan, Winnipeg and Clearwater

Added	Count	Notes
Existing posts moved to the right river	16	Fort Edmonton, Fort Augustus, Fort Pitt, Fort George (Saskatchewan), Fort Carlton, Fort Battleford and Rocky Mountain House to the North Saskatchewan; Batoche to the South Saskatchewan; Fort Liard to the Liard; Bas-de-la-Rivière, Slave Falls Portage and Rat Portage to the Winnipeg River; Fort Lac La Pluie to the Rainy River; Fort Qu'Appelle to the Qu'Appelle; Fort St. James to Stuart Lake; Fort Témiscamingue to Lake Timiskaming (backup records every old waterway id)
Explorers	10	Joseph and Thomas Frobisher, James Finlay, Joseph Howse, John Rowand, Robert Campbell, John McLeod, Alexander Caulfield Anderson, Henry Youle Hind, James Hector. Lifetime portraits for five (Frobisher by Dulongpré c. 1800; photographs of Campbell, Anderson, Hind and Hector)
Explorer–waterway links	93	Henday, Cocking, Tomison, Pond, Thompson, Fidler, Franklin, Palliser and the La Vérendryes on the new rivers; every new explorer has at least one
Locations	16	Fort Smith, Fort Halkett, Fort St. Pierre, Fort Dauphin, South Branch House, Chesterfield House, Hudson House, Buckingham House, Fort Calgary, Blackfoot Crossing, Fort Espérance, Fort Charlotte, Île-à-la-Crosse, Mackenzie Rock, Sainte-Marie among the Hurons, Penetanguishene. Five have no image (nothing suitable and licensed on Commons): Fort Halkett, South Branch House, Hudson House, Île-à-la-Crosse, Mackenzie Rock
Notable figures	6	Maskepetoon, Payipwat (Piapot), Mistawasis, Ahtahkakoop, Jerry Potts, Pierre Falcon. Lifetime portraits for four: Piapot (photograph c. 1885), Mistawasis and Ahtahkakoop (the 1886 photograph of four Cree chiefs), Potts (photograph)
Timeline events	37 new; 12 existing unlinked timeline events given their links (HBC charter, Thanadelthur's mission, Cumberland House with Hearne, Fraser's descent, Thompson's arrival on the Pacific, Seven Oaks, Red River Resistance, Manitoba, Northwest Resistance with Batoche, Lake of the Woods massacre, and others)	
What Happened Here events	15 new; 14 existing unlinked events given links (Treaty 6 at Fort Carlton, Seven Oaks, Selkirk settlers, Fort Rouge, Treaty 1, the rendezvous at Grand Portage and Fort William, Fort St. Charles, HBC founded, NWC formed, Plains of Abraham, Thompson's Mandan journey, and others)	

Added	Count	Notes
Pronunciations	18	Cree, Blackfoot, Dene, Dakelh, Sekani, Nuxalk, Kaska, Anishinaabe, Nakoda and Wendat terms, and pemmican
Quiz	1	"Rivers of the Fur Trade Interior", 8 questions, grade 7-9
Deep Dive / lesson plan	1	"River Highways: How Indigenous Routes Became the Fur Trade's Roads", grade 4-6, Indigenous Routes
Field trip	1	"Sainte-Marie among the Hurons, 1639-1649", grade 4-6, six stops
Reading-guide entries	119	Dictionary of Canadian Biography, Canadian Encyclopedia, Parks Canada, Hudson's Bay Company Archives and Wikipedia, every URL verified
Images	88 URLs	Commons, public domain or CC BY / CC BY-SA, credited; includes Franklin's 1828 plate of the Clearwater valley from Methye Portage, Pond's 1785 map, Fidler's copy of Ac ko mok ki's 1801 map, Thompson's 1814 map, Palliser's general map, Fort Carlton in 1877, Back's buffalo pound, the 1878 photograph of the police at Fort Calgary, and the 1908 water-power plan of the Winnipeg River

Added the same day (`2026-10-17-primary-sources-interior.ts`): 14 primary-source pointer documents (Brébeuf's and Ragueneau's Relations, the La Vérendrye journals, Henday, Cocking, Pond, Henry the Younger, Fidler and the Ac ko mok ki map, Hind, the Palliser reports, Anderson's gold-rush handbook, Morris's Treaties with Erasmus's Buffalo Days and Nights, Falcon's song, Mackenzie's 1793 journal). After the pack, every record in Turso has at least two kinds of source among its reading guides (`2026-10-17-reading-guide-gaps.ts`, 172 guides in two passes), and no record is without one.

9.10 Content pack 5, The Bay and the Barren Lands (applied 2026-10-17)

Data in `backend/prisma/turso-migrations/packs/bay-and-barren-lands.ts`, applied by `2026-10-17-pack-bay-and-barren-lands.ts` (dry run, backup, apply, verify; idempotent). Scope: the rivers of Hudson Bay and James Bay that the earlier packs left as bare names or did not have at all: the Omushkego Cree rivers of the Ontario lowlands, the Eeyou Istchee rivers of the Quebec side, Lake Mistassini and Lake Abitibi on the overland routes, the Nunavik rivers to Ungava Bay, and the Sayisi Dene and Caribou Inuit rivers of the Barren Grounds to Baker Lake. People come from the Dictionary of Canadian Biography wherever it has them (thirteen explorers and all six figures); Low, Pike, Seton and J. W. Tyrrell, whom it does not cover, rest on Wikipedia, the Geological Survey reports and their own books, and every date in the pack was checked against those sources before it was written.

Added	Count	Notes
Waterways	18	Severn, Winisk, Attawapiskat, Rupert, Eastmain, Nottaway, Harricana, La Grande, Great Whale, Seal, Kazan, Dubawnt, Baker Lake, Ungava Bay, Koksoak, George, Lake Mistassini, Lake Abitibi. OpenStreetMap geometry for seventeen (Lake Mistassini thinned from 114,000 to 39,000 points; the Quebec rivers exported under their French OSM names); the Eastmain has no named OSM relation and stays a pin. Fur-trade rows for the Severn, Rupert, Eastmain and Koksoak

Added	Count	Notes
Existing posts moved to the right river	3	York Factory to the Hayes (the Hayes had no post and no explorer), Fort Severn to the Severn, Rupert House to the Rupert; the backup records the old waterway ids
Explorers	17	Albanel, Isham, Coats, Mitchell, Isbister, Graham, Wales, Erlandson, Finlayson, McLean, Bell, Low, Pike, Hanbury, Hornby, J. W. Tyrrell, Seton. Lifetime portraits for two only (Seton, Harris and Ewing photograph; Bell, the photograph in the Royal Society of Canada's Proceedings, dated no later than 1917). Commons has no lifetime likeness of Hornby, Hanbury, McLean, Pike, Low, Nickawa or Christian, and none exists for the eighteenth-century men
Explorer–waterway links	74	Hearne on the Seal, Kazan, Dubawnt and Thelon; James and Foxe at the Severn; Groseilliers at the Rupert; de Troyes and Iberville on the Albany, Rupert, Moose, Abitibi, Severn and Hayes; Radisson and Kelsey on the Hayes; J. B. Tyrrell on the Dubawnt, Kazan, Baker Lake and Chesterfield Inlet; every new explorer has at least one. The Albany and the Hayes, which had none, now have four and four
Locations	12	Henley House, Eastmain House, Fort George (Chisasibi), Fort Chimo, Fort Severight, Mistassini Post, Fort Richmond, Great Whale River Post, Baker Lake Post, Hornby Point, Winisk (Peawanuck), Attawapiskat. Fort Richmond has no image
Notable figures	6	Miscomote, Wappisis, Abishabis, Inukjuarjuk, Zhauwuno-geezhigo-gaubow (Jack Fiddler), Frances Nickawa; all from the Dictionary of Canadian Biography. No lifetime likeness on Commons for any (Forster's 1921 portrait of Nickawa is not there)
Timeline events	33 new; the unlinked "Thanadelthur Captured by Cree Raiders" linked to Hudson Bay and York Factory	
What Happened Here events	10 new (Albanel at the Rupert, the taking of Fort Albany, the surrender of York Fort, the Albany hangings, the transit of Venus, Erlandson at Lake Melville, the death of Abishabis, the Tyrrells on the coast, Jack Fiddler at Norway House, the last entry at Hornby Point)	
Pronunciations	22	Omushkego and Eeyou Cree, Inuktitut, Naskapi, Dene and Algonquin place names and nations, and windigo
Quiz	1	"The Bay and the Barren Lands", 8 questions, grade 7-9
Deep Dive / lesson plan	1	"The Bay: Cree, Inuit, Dene and the Company on Hudson Bay", grade 7-9, Fur Trade
Field trip	1	"The Thelon: Hornby Point and the Barren Lands", grade 7-9, six stops

Added	Count	Notes
Reading-guide entries	151	Dictionary of Canadian Biography, Canadian Encyclopedia, Hudson's Bay Company Archives, Internet Archive editions of the sources, Parks Canada and Wikipedia; every URL verified (five Commons image URLs answered only 429 on the final check and were accepted on the strength of their file records)
Images	78 URLs	Commons, public domain or CC BY / CC BY-SA, credited; includes Andrew Graham's 1768-70 plan of the bay, Moses Norton's 1760 draught, Pierre Allemand's chart of the forts taken in 1686, Low's map of Lake Mistassini, the 1935 Seal River survey sheet, the Keewatin survey of the Winisk, the stove in Hornby's cabin from the NWT Archives, and the 1937 Hoare and Clarke photograph on the Thelon

Added the same day (`2026-10-17-primary-sources-bay.ts`): 17 primary-source pointer documents (Albanel's Relation of 1672, Isham's and Graham's Observations, Coats's Geography, Wales's Churchill journal, the Severn journal on Abishabis, Erlandson's crossing, McLean's Notes, Turner's Ethnology of Ungava, Bell's Attawapiskat report, Low's Labrador reports, Tyrrell's Dubawnt and Kazan report with his brother's book, Pike, Hanbury, Seton, Christian's diary, and Killing the Shamen).

9.11 Content pack 6, The Yukon (applied 2026-10-17)

Data in `backend/prisma/turso-migrations/packs/yukon.ts`, applied by `2026-10-17-pack-yukon.ts` (dry run, backup, apply, verify; idempotent). Scope: the Yukon, which existed as a row with no explorers and one post, and everything around it: the tributaries and headwater lakes, the two routes by which the Hudson's Bay Company reached the river from the Mackenzie in the 1840s, the surveys of the 1880s, the Klondike, and the Hän, Tagish and Ta'an Kwäch'än response to it. The Yukon row itself went from no explorers and one post to 17 explorers and 7 posts and sites. People rest on the Dictionary of Canadian Biography where it has them (ten explorers, four figures, each entry read in full) and on Wikipedia and their own books for the Americans it does not (McQuesten, Harper, Ladue, Carmack, Ogilvie, Kennicott, Zagoskin, Jim Boss).

Added	Count	Notes
Waterways	16	Pelly, Stewart, Klondike, Teslin River, White, Porcupine, Peel, Dease River, Fortymile, Lake Laberge, Bennett Lake, Tagish Lake, Kluane Lake, Teslin Lake, Atlin Lake, Frances Lake; OpenStreetMap geometry for all sixteen. Fur-trade rows for the Pelly and the Porcupine
Explorers	17	Bell, Murray, Hardisty, Schwatka, Whymper, Dawson, Bompas, Constantine, Steele, Zagoskin, Kennicott, McQuesten, Harper, Ogilvie, Carmack, Ladue, and Campbell's links extended. Lifetime likenesses for eight (Schwatka, 1881 engraving; Dawson, Notman photograph 1892; Bompas; Steele, 1900; Zagoskin; McQuesten; Ogilvie; Harper in Farciot's 1884 group at Fort Yukon; Carmack by Woodside c. 1900). None exists on Commons for Bell, Murray, Hardisty, Whymper, Constantine, Kennicott (an engraving of uncertain date was passed over) or Ladue
Explorer–waterway links	56	Campbell on the Frances, Pelly, Yukon, Stewart and Dease; Bell on the Peel, Porcupine and Yukon; Franklin on the Peel; the Yukon row goes from no explorers to seventeen
Locations	15	Fort Selkirk, Frances Lake Post, Lapierre House, Fort Reliance (Yukon), Forty Mile, Fort Constantine, Tr'ochëk, Moosehide, Dawson City, Bonanza Creek Discovery Claim, Carcross, Bennett, Chilkoot Pass, Miles Canyon, Five Finger Rapids. Three have no image (Frances Lake Post, Fort Reliance, Fort Constantine)

Added	Count	Notes
Notable figures	5	Keish (Skookum Jim), Káa Goox (Dawson Charlie), Shaaw Tláa (Kate Carmack), Isaac, Jim Boss. Lifetime photographs for four (Keish by Duclos, Dawson, 1898; Káa Goox and Carmack from Library and Archives Canada; Shaaw Tláa; Jim Boss from the Parks Canada designation record); none on Commons for Chief Isaac
Timeline events	30	From McLeod on the Dease (1831) to the World Heritage inscription (2023)
What Happened Here events	8	Campbell's walk to the Pelly, Fort Yukon in Russian territory, the sack of Fort Selkirk, Schwatka in Miles Canyon, gold on Rabbit Creek, the Hän leaving Tr'ochëk, Steele's rules at Bennett, Keish's return to Carcross
Pronunciations	21	Hän, Northern and Southern Tutchone, Gwich'in, Tlingit, Tagish and Kaska names
Quiz	1	"The Yukon: River, Company and Klondike", 8 questions, grade 7-9
Deep Dive / lesson plan	1	"The Yukon: A River Road from the Company to the Klondike", grade 7-9
Field trip	1	"Klondike, 1896-1898: From Tr'ochëk to Dawson", grade 7-9, six stops
Reading-guide entries	135	Dictionary of Canadian Biography, Canadian Encyclopedia, Hudson's Bay Company Archives, Parks Canada, Internet Archive editions and Wikipedia; every URL verified, with the Canadian Encyclopedia slugs checked individually (it has no articles on most of the Yukon's rivers)
Images	76 URLs	Commons, public domain or CC BY / CC BY-SA, credited; includes Isbister's 1845 sketch map of the Peel, Marsh Lake and Forty Mile about 1899, the 1899 Bonanza Creek drawing, and a 1897 fortune-seekers' guide

Added the same day (`2026-10-17-primary-sources-yukon.ts`): 15 primary-source pointer documents (Campbell's Discovery and Exploration of the Youcon, Murray's Journal of the Yukon, Hardisty's Loucheux Indians, Whymper with Dall, Schwatka's book and report, Dawson's 1887 report, Ogilvie's 1897 pamphlet and Early Days, Hayne's Pioneers of the Klondyke, Adney's Klondike Stampede, Steele's Forty Years, Haskell's Two Years, Carmack's My Experiences, Zagoskin's Travels, Martha Black's My Seventy Years, and Angela Sidney's Tagish account of the discovery).

Owner's review of 2026-10-17, to be taken up after this pack: the Parsnip River has no fur-trade record or posts, and the app has no waterway-to-waterway relation at all (the Peace, Finlay and Parsnip are joined only in prose); the Kaministiquia route (Dog Lake, Dog River, Lac des Mille Lacs) is absent though Great Dog Portage exists; and a full sweep of every template field for every record is required once the packs are done. Measured on 2026-10-17 across 150 waterways: 118 without a fur-trade record, 111 with no connected notable figure, 144 with no archaeological discovery, 81 without video, 64 without a post or site, 55 without an Indigenous name, 16 without an explorer, 9 without geometry.

9.12 Gap pack: the Kaministiquia route and the Parsnip (applied 2026-10-18)

From the owner's review of 2026-10-17. Data in `packs/kaministiquia-and-parsnip.ts`, applied by `2026-10-18-gap-kaministiquia-parsnip.ts`, a copy of the pack machinery with the lesson and field trip made optional and a `furTradeForExisting` list for rows that lacked a fur-trade record.

Added or changed	Notes
Waterways (6)	Dog Lake, Dog River, Lac des Mille Lacs, Savanne River (no OpenStreetMap geometry; a pin), Pack River, McLeod Lake; geometry for the other five
Fur-trade records (4)	Parsnip River and Kaministiquia River (existing rows that had none), Dog Lake and McLeod Lake

Added or changed	Notes
Explorer	Zacharie Robutel de La Noue, who built Fort Kaministiquia in 1717 (Dictionary of Canadian Biography)
Links (8)	de Noyon on Dog Lake, Dog River and Lac des Mille Lacs (1688); La Noue on the Kaministiquia and Lake Superior; Dulhut on Lake Superior; Fraser and Stuart on the Pack and McLeod Lake (1805); Mackenzie on the Peace (1793)
Location	Kakabeka Falls, the Mountain Portage of the Kaministiquia route
Posts moved	Fort Kaministiquia from the Lake Superior row to the Kaministiquia; Fort McLeod (BC) from the Peace to McLeod Lake, which the Parsnip and the Pack reach
Events, pronunciations, quiz	3 timeline events, 2 What Happened Here events (La Noue's post; Mackenzie at Finlay Forks), 3 pronunciations, a four-question quiz; 20 reading guides

Still open from the same review: the data model has no waterway-to-waterway relation, so the Parsnip, Finlay and Peace are joined only in prose. A tributary or confluence relation is proposed as a code change for the owner's approval before the template-field sweep (section 9.11, last paragraph) begins.

9.13 Content pack 7, The Atlantic (applied 2026-10-18)

Atlantic Canada was the thinnest part of the data (Newfoundland and Labrador had three waterways, New Brunswick one). Data in `packs/atlantic.ts`, applied by `2026-10-18-pack-atlantic.ts`; primary sources by `2026-10-18-primary-sources-atlantic.ts`. The Gulf of St. Lawrence, Newfoundland, Labrador and the Maritimes from the Bristol and Portuguese voyages to the Admiralty surveys, with the Mi'kmaq, Wolastoqiyik, Beothuk, Innu and Labrador Inuit at the centre. People rest on the Dictionary of Canadian Biography (34 biographies read in full) and on Wikipedia for Mina Hubbard and Nonosabasut.

Added or changed	Notes
Waterways (21)	Gulf of St. Lawrence (no single OpenStreetMap geometry; a pin at 48°N 62°W), Exploits River, Beothuk Lake (Red Indian Lake), Notre Dame Bay, Trinity Bay, Conception Bay, Humber River (Newfoundland; no OSM geometry, a pin), Hamilton Inlet (with Lake Melville's geometry), Churchill River (Labrador), Naskaupi River, Sandwich Bay, Nain Bay, Miramichi, Restigouche, Annapolis Basin, Minas Basin, Northumberland Strait, Cabot Strait, Bras d'Or Lake, Chignecto Bay; 19 with geometry
Fur-trade records (4)	Hamilton Inlet (Fornel 1743, Hudson's Bay Company 1836), Sandwich Bay (Cartwright 1775), Annapolis Basin (de Monts to d'Aulnay), Bras d'Or Lake (Denys 1654-69)
Explorers (28)	Corte-Real, Verrazzano, Roberval, Jean Alfonse, Gilbert, Edward Hayes, John Guy, John Mason, Whitbourne, Kirke, de Monts, Poutrincourt, Lescarbot, La Tour, Razilly, Denys, Argall, Fornel, Laure, Palliser, Cartwright, Jens Haven, Buchan, Cormack, Jukes, Bayfield, DesBarres, Mina Benson Hubbard
Portraits (8)	Lifetime likenesses only: Roberval (Clouet sketch, c. 1540), Palliser (attr. Dance, c. 1775), Cartwright (Hilton, c. 1791), DesBarres (credited to Peachey), Jukes (photograph c. 1860), Mina Hubbard (1905), and the figures Demasduit (Lady Hamilton, 1819) and Mikak (Russell, 1769). No lifetime likeness could be confirmed for the other twenty; they are left without portraits rather than given a later imagining
Notable figures (4)	Demasduit, Nonosabasut, Mikak, Tuglavina (Shawnadithit and Membertou already existed)

Added or changed	Notes
Links (66)	Including the existing Cabot, Cartier, Champlain, Cook, Jolliet (the 1694 Labrador voyage), John Davis (Hamilton Inlet, 1586), d'Iberville (Placentia, 1696), McLean and Erlandson (the Grand Falls, 1839) and A. P. Low (1894); new links on the existing Strait of Belle Isle, Placentia Bay, Baie des Chaleurs, Saguenay, Saint John, Bay of Fundy, St. Lawrence and George River rows
Locations (14)	Cupids, Trinity, Boyd's Cove Beothuk Site, Placentia (Plaisance), North West River, Rigolet, Cartwright, Nain Moravian Mission, Fort La Tour, Grand-Pré, Fort Beauséjour, Saint-Pierre (St. Peter's), Nipisiguit, Miscou Post
Post moved	Port-Royal (Habitation) from the Bay of Fundy row to the new Annapolis Basin
Events, pronunciations	37 timeline events, 10 What Happened Here events (Gilbert at St. John's; Sainte-Croix; the Théâtre de Neptune; Bull Arm; Fornel; Chateau Bay; Buchan's marines; the taking of Demasduit; Shawnadithit's drawings; the Grand Falls), 26 pronunciations (Mi'kmaq, Wolastoqey, Beothuk, Innu-aimun, Inuttitut)
Quiz, lesson, field trip	Quiz "The Atlantic Gateway" (8 questions); lesson "The Atlantic Gateway: Where Europe First Met the Northeast" (grades 7-9); field trip "Port-Royal, 1606-1607: The Winter of Good Cheer" (grades 4-6, six stops)
Primary sources (16)	The Cabot and Corte-Real letters (Biggar), Hayes on Gilbert (Hakluyt), Guy's letters (Purchas, Cell), Mason's and Whitbourne's Discourses, Champlain's and Lescarbot's Acadia, Biard's Relation of 1616, Denys (Ganong), Fornel's Relation, Cartwright's Journal, Cormack's Narrative, Shawnadithit's drawings and the Beothuk record (Howley), Jukes, Bayfield's Sailing Directions, Mina Hubbard; every Internet Archive identifier checked

Rules kept: every statement was read in the Dictionary of Canadian Biography or Wikipedia during authoring, and details that could not be found there were cut (Cartier's return through the Cabot Strait, the Fornel page numbers, Jukes's later career, Bayfield's exact end date). The Beothuk material follows the DCB entries on Demasduwit and Shawnadithit closely. Verify after apply: 192 waterways, 187 explorers, 618 explorer-waterway links, 199 locations, 65 notable figures, 244 timeline events, 132 What Happened Here events, 172 pronunciations, 15 quizzes, 33 lessons, 13 field trips, 2,002 reading guides, 176 primary-source documents (85 for grades 10-12, 80 for 7-9, 11 for 4-6). Gulf of St. Lawrence 6 explorers; Annapolis Basin 7 explorers and Port-Royal; Hamilton Inlet 5 explorers, North West River and Rigolet. The production API served all 192 waterways, 187 explorers and 65 figures within a minute of the apply. Three timeline and four What Happened Here events from older packs remain unlinked and go to the sweep.

9.4 Document grade bands (approved and applied 2026-10-17)

Proposed from a Flesch-Kincaid estimate of each transcription (≤ 7 → 4-6, ≤ 11 → 7-9, above → 10-12), checked against the fourteen documents that already had a band (their estimates run 4.4 to 10.9 for 7-9 and 5.6 to 9.3 for 4-6). Result: 5 at 4-6 (Champlain's founding of Quebec and his Hochelaga description, Cartier on scurvy, Joutel on La Salle's last expedition, Fidler with the Peigan), 19 at 7-9, 10 at 10-12 (La Salle's Louisiana claim, Tyrrell's bone beds, Back's three, Turnor's instructions, Norton's instructions and trade report, Stuart's peace mission report, La Pérouse's attack account). The full list is in the script's `GRADES` map.